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Cancer Biology

Multi-gradient Permutation Survival Analysis Identifies Mitosis and Immune Signatures Steadily Associated with Cancer Patient Prognosis

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eLife Assessment

This paper contains **valuable** ideas for methodology concerned with the identification of genes associated with disease prognosis in a broad range of cancers. However, there are concerns that the statistical properties of MEMORY are **incompletely** investigated and described. Further, more precise details about the implementation of the method would increase the replicability of the findings by other researchers.

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Abstract

The inconsistency of the association between genes and cancer prognosis is often attributed to many variables that contribute to patient survival. Whether there exist the Genes Steadily Associated with Prognosis (GEARs) and what their functions are remain largely elusive. Here we developed a novel method named "Multi-gradient Permutation Survival Analysis " (MEMORY) to screen the GEARs by using RNA-seq data from the TCGA database. We employed a network construction approach to identify hub genes from GEARs, and utilized them for cancer classification. In the case of lung adenocarcinoma (LUAD), the GEARs were found to be related to mitosis. Our analysis suggested that LUAD cell lines carrying *PIK3CA* mutations exhibit increased drug resistance. For breast invasive carcinoma (BRCA), the GEARs were related to immunity. Further analysis revealed that *CDH1* mutation might regulate immune infiltration through the EMT process. Moreover, we explored the prognostic relevance of

mitosis and immunity through their respective scores and demonstrated it as valuable biomarkers for predicting patient prognosis. In summary, our study offered significant biological insights into GEARs and highlights their potentials as robust prognostic indicators across diverse cancer types.

Introduction

Cancer is one of the leading causes of death worldwide, with over 200 types identified. Despite major advances in the field, accurately predicting cancer patient prognosis remains a significant challenge[1,2]. Previous studies have highlighted the complexity of this task, which is influenced by various factors, including cancer types, clinical stages, therapeutic interventions, nursing care, unexpected comorbidities, and other non-cancer related illnesses that may interplay[3–6]. Furthermore, the gene expression plays a crucial role in predicting cancer patient prognosis, such as *HER2*, *VEGF*, *Ki67*, etc[7–12]. Nevertheless, there is still a need for further improvement in prognostic accuracy to better inform treatment decisions and patient outcomes.

Survival analysis is commonly used to assess the correlation between genes and cancer patient prognosis[13]. However, inconsistent findings are frequently observed, even within the same type of cancer. For instance, studies exploring the correlation between *CCND1* and prognosis in non-small cell lung cancer (NSCLC) reported contrasting results, including positive correlation, inverse correlation, or negligible influence[14–16]. These discrepancies can be attributed to various influential factors, such as differences in sample size, cohort characteristics (including cancer subtypes and tumor staging) and variations in therapeutic approaches[17]. Such observations raise an important question: whether there exist Genes Steadily Associated with Prognosis (hereafter referred to as GEARs) that correlate with patient outcomes across different conditions, particularly considering the varying sample sizes used in different studies? Affirmative answers to this question would have significant implication not only for the development of more accurate prognostic models, but also for enhancing our understanding of cancer biology.

The Cancer Genome Atlas (TCGA) is a comprehensive cancer genomics project initiated in the United States. It consists of transcriptome data, genomic data, and clinical information pertaining to 33 different cancer types, making it the largest cancer clinical sample database currently available. In this study, we developed a novel method named "Multi-gradient Permutation Survival Analysis" (MEMORY) with the utilization of TCGA RNA-seq database, and assessed the potential existence of GEARs across 15 cancer types, each comprising a cohort of over 200 patients. Furthermore, we evaluated the prognostic predictive power of these GEARs and explored their potential biological functions in driving cancer progression.

Results

Multi-gradient permutation survival analysis identifies GEARs associated with mitosis and immune across multiple cancer types

GEARs are a group of genes consistently and significantly correlate with cancer patient survival, independent of the sample size. To identify these GEARs, we developed a novel method called "Multi-gradient Permutation Survival Analysis " (MEMORY). This method allows us to assess the correlation between a specific gene and cancer patient prognosis using available transcriptomic datasets (**Figure 1A**, Table 1). Initially, we started with a sampling number at 10% of the cohort size, then gradually increased the sampling number with each 10% increment, which was analyzed with 1000 permutations. By calculating the statistical probability of each gene's association with patient survival, we can identify a group of GEARs for further analyses.

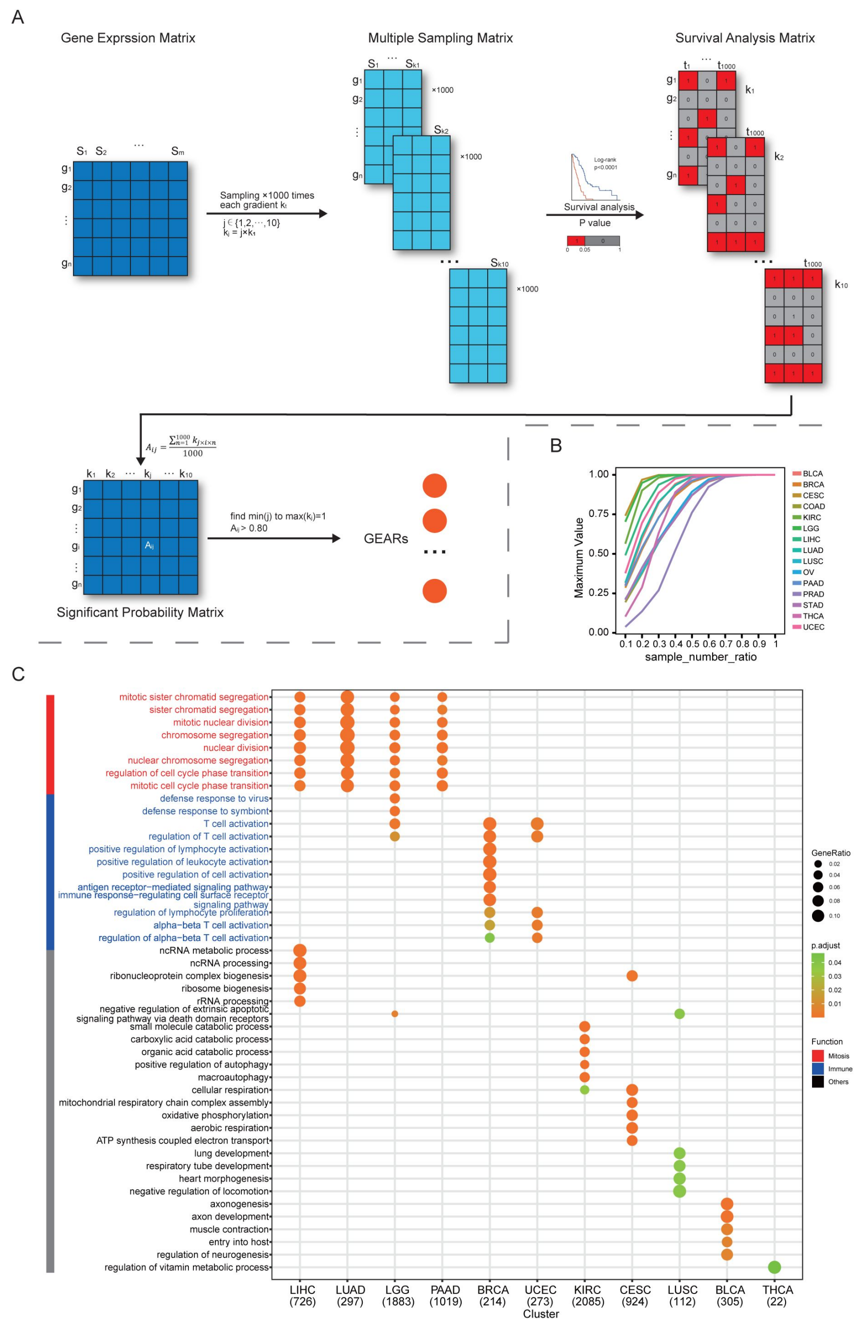


Figure 1

MEMORY uncovers the enrichment of mitosis and immune signatures in multiple cancers.

(**A**) Algorithm of MEMORY. Sample sizes, ranging from 10% to 100% with 10% intervals (“gradient”), were used for 1000 permutations of survival analyses of all 15 cancer types. Each matrix was divided into high and low expression groups based on median gene expression values, and survival analyses were performed. Log-rank test significance results were coded as 1 for significant ($P < 0.05$) and 0 for non-significant ($P > 0.05$) outcomes, forming survival analysis matrices and the summarized significant probability matrix, which allowed the identification of GEARs. (**B**) The maximum of significant probability for each gradient sample size of every cancer type. Sample number rate refers to the percentage of samples in each sampling gradient compared to the total number of samples. (**C**) The pathways were enriched by GOEA based on the GEARs of each cancer type. The displayed pathways represent the top 5 most significant pathways for each cancer type. Mitosis-related pathways were marked in red whereas immune-related pathways were marked in blue. Only pathways with a Benjamini–Hochberg–adjusted $p < 0.05$ are displayed.

In this study, we utilized the TCGA datasets for several reasons, including the diversity of cancer types, the availability of gene expression profiles and patient prognosis information. We set a minimum cohort size of 200 and included 15 eligible cancer types for analysis. These cancer types include bladder urothelial carcinoma (BLCA), breast invasive carcinoma (BRCA), cervical squamous cell carcinoma and endocervical adenocarcinoma (CESC), colon adenocarcinoma (COAD), kidney renal papillary cell carcinoma (KIRC), brain lower grade glioma (LGG), liver hepatocellular carcinoma (LIHC), lung adenocarcinoma (LUAD), lung squamous cell carcinoma (LUSC), ovarian serous cystadenocarcinoma (OV), pancreatic adenocarcinoma (PAAD), prostate adenocarcinoma (PRAD), stomach adenocarcinoma (STAD), thyroid carcinoma (THCA), and uterine corpus endometrial carcinoma (UCEC). As the number of samples increases, the significant probability of certain genes gradually approaches 1. Once this score reaches 0.8, previously employed as an empirical standard for survival analyses, and remained above this value consistent with further sample gradient increase, the gene is considered a GEAR (**Figure 1B**)[18]. Remarkably, we successfully identified a set of GEARs across all 15 cancer types (**Supplementary Figure 1-2**, Table 2). The GEAR counts in most cancer types ranged from 100 to1000, with exceptions of CESC, KIRC, LGG, and PAAD with over 1000 GEARs, and THCA with only 22 GEARS (Table 2). In LUAD, the top 10 GEARs with the highest significance probabilities were *TLE1*, *GNG7*, *ERO1A*, *ANLN*, *DKK1*, *TMEM125*, *S100A16*, *KNL1*, *STEAP1*, and *BEX4*. Most of these genes are known to promote LUAD malignant progression except for S100A16[19–27]. In other cancer types, *BEX4* was identified as a common GEAR in KIRC, LGG, PAAD, and STAD (Table 2). *BEX4* is reported as a proto-oncogene promoting cancer onset and malignant progression in multiple cancers, including LUAD, glioblastoma multiforme and oral squamous cell carcinoma[26,28,29]. We also identified the top 1 GEAR in individual cancer types, such as *TLL1* (BLCA), *PGK1* (BRCA), *RFXANK* (CESC), *DPP7* (COAD), *VWA8* (KIRC), *SCMH1* (LGG), *HILPDA* (LIHC), *TLE1* (LUAD), *CD151* (LUSC), *ANKRD13A* (OV), *SOCS2* (PAAD), *DRG2* (PRAD), *ADAMTS6* (STAD), *PSMB8* (THCA), *ASS1* (UCEC) (Table 3). Many of these genes were previously reported to be associate with tumorigenesis[19,30–41]. For example, *TLE1* is known as a transcriptional repressor that promotes cell proliferation, migration, and inhibits apoptosis in LUAD[42,43]. Additionally, *PGK1*, a key enzyme in the glycolytic process, has been shown to promote cell proliferation, migration, and invasion in multiple cancers[44,45]. These findings demonstrate the intricate link between the functionality of GEARs and the initiation and progression of cancer.

To gain deep insights into the biological functions of these identified GEARs, we conducted Gene Ontology (GO) enrichment analysis (**Figure 1C**). Interestingly, we found the mitosis-related pathways were enriched in LICH, LUAD, LGG, and PAAD, and the immune-related pathways were enriched in BRCA and UCEC[46]. Additionally, other cancer types exhibited enrichment in various pathways, such as organic acid metabolism pathway in KIRC, oxidative phosphorylation pathway in CESC, organ development-related pathways in LUSC, neurogenesis-related pathways in BLCA, and sustenance metabolism pathways in THCA. Given the crucial role of mitosis in cancer progression and the significance of the immune system in cancer-host interactions, we specifically focused on the mitosis and immune-related GERAs, which accounted for approximately 40% of all the analyzed cancers.

Identification of hub genes in mitosis and immune-related cancers

To identify crucial genes within the GEARs, we conducted and extracted the higher-ranked edges to construct the core survival network (CSN) (**Figure 2A**, Table 4). The top 10 GEARs with the highest degree in these networks, selected from those with mean expression >10 TPM, were defined as hub genes[47]. We then classified the samples using the hub genes derived from these networks and evaluated their clinical relevance (**Supplementary Figure 3A-O**, **Supplementary Figure 4A-O**). A significant correlation with cancer stages (TNM stages) is observed in most cancer types except for LUSC, THCA and PAAD (**Supplementary Figure 5A-K**). Furthermore, we conducted GO enrichment analysis on the hub genes selected from CSNs and found that the results were consistent with the GEARs analysis (Table 5). Specifically, the hub genes in LIHC, LGG, and LUAD were enriched in mitosis-related pathways, whereas the hub genes in BRCA and UCEC were enriched in immune-related pathways (**Figure 2B**). For instance, the 9 hub genes in LUAD were associated with functions related to mitosis, whereas the 8 hub genes of BRCA were associated with immune-related functions (**Figure 2C-D**).

Next, we conducted survival-dependent analyses on the hub genes of LGG, LIHC, and LUAD. These analyses revealed that mitosis-related hub genes were closely associated with cancer cell viability, especially those hub genes that were correlated with multiple cancer types, such as *CDC20*, *TOP2A*, *BIRC5* and *TPX2*[48–51] (**Supplementary Figure 6A-C**). The importance of these genes is well established. For instance, *TPX2*, a hub gene in all three cancers, is known to play a crucial role in normal spindle assembly during mitosis and is essential for cell proliferation[52]. The significance of these hub genes for the survival of cancer cells suggests that the expression levels of these hub genes could be used for inhibitors screening of tumor growth. By integrating the Genomics of Drug Sensitivity in Cancer (GDSC) and Connectivity Map (cMAP) databases, we identified 76 individual compounds that were able to effectively suppress the expression of the 10 hub genes in LUAD cell lines (**Figure 2E**, **Supplementary Figure 6D-E**). These compounds also significantly inhibited LUAD cell growth and might serve as potential therapeutic agents for LUAD.

We then focused on our analysis on LUAD dataset, which had a larger sample size compared to LGG and LIHC. According to the expression of the hub genes, we classified the LUAD samples into three subgroups: mitosis low (ML), mitosis medium (MM), and mitosis high (MH) (**Figure 2F**, Table 6). Previous study has reported three classic subgroups of LUAD known as the terminal respiratory unit (TRU), the proximal-proliferative (PP), and the proximal-inflammatory (PI) subgroups[53]. Our analysis revealed that the ML subgroup was primarily enriched with TRU subgroup and a small number of samples from the PP subgroup, but not the PI subgroup. The MH subgroup showed high expression of mitosis-related genes and predominantly encompassed PI and PP subgroups, whereas the MM group exhibited intermediate levels of mitosis-related gene expression (**Figure 2F-G**). This suggests that the new categorization method may help identify new factors that influence patient prognosis (**Supplementary Figure 6F-G**).

Distinct genetic mutation landscapes characterize different clusters of LUAD

The expression of hub genes provided crucial indicators for distinguishing various subgroups of LUAD. However, the underlying mechanisms driving these variations in expression patterns remain unknown. In our subsequent research, we aimed to explore the genomic differences, particularly in terms of gene mutations, among different LUAD subgroups. Initially, we analyzed the variations of genetic mutation landscapes of different LUAD subgroups by assessing the tumor mutation burden (TMB). Interestingly, we found significant differences in TMB among the various subgroups of LUAD (**Figure 3A**). Common driver genes were heterogeneously distributed across the three molecular subgroups. *ALK* and *ROS1* fusions were most enriched in the prognosis-favorable ML and MM clusters, while *KRAS*, *EGFR*, *BRAF* and *ERBB2* mutations were more evenly distributed or enriched in the less prognosis-favorable MH group (**Figure 3B**). Due to limited counts for some fusion events, formal statistical comparison was not performed, but the distributional patterns suggested distinct subtype preferences across different driver events. Moreover, low tumor mitotic activity was associated with better prognosis in LUAD subgroups with *EGFR* mutations or pan-negative (no oncogenic alteration in genes including *KRAS*, *EGFR*, *BRAF*, *ERBB2*, *PIK3CA*, *ALK* and *ROS1*) but not in those with *KRAS* or *BRAF* mutations (**Figure 3C-D**, **Supplementary Figure 6H-I**). Besides, we observed unique mutation characteristics in each of these three subgroups (**Figure 3E-G**). For example, *TP53* mutation, a prevalent tumor suppressor gene mutation, was frequently observed in the MH subgroup with a mutation rate of 73%, compared to mutation rates of 14% in the ML subgroup and 39% in the MM subgroup. While genes, such as *CSMD3*, *RP1L1* and *ZFHX4,* exhibited similar trend in mutation frequency across the three subgroups. These findings indicate that substantial genomic differences among these three LUAD subgroups are based on the hub genes.

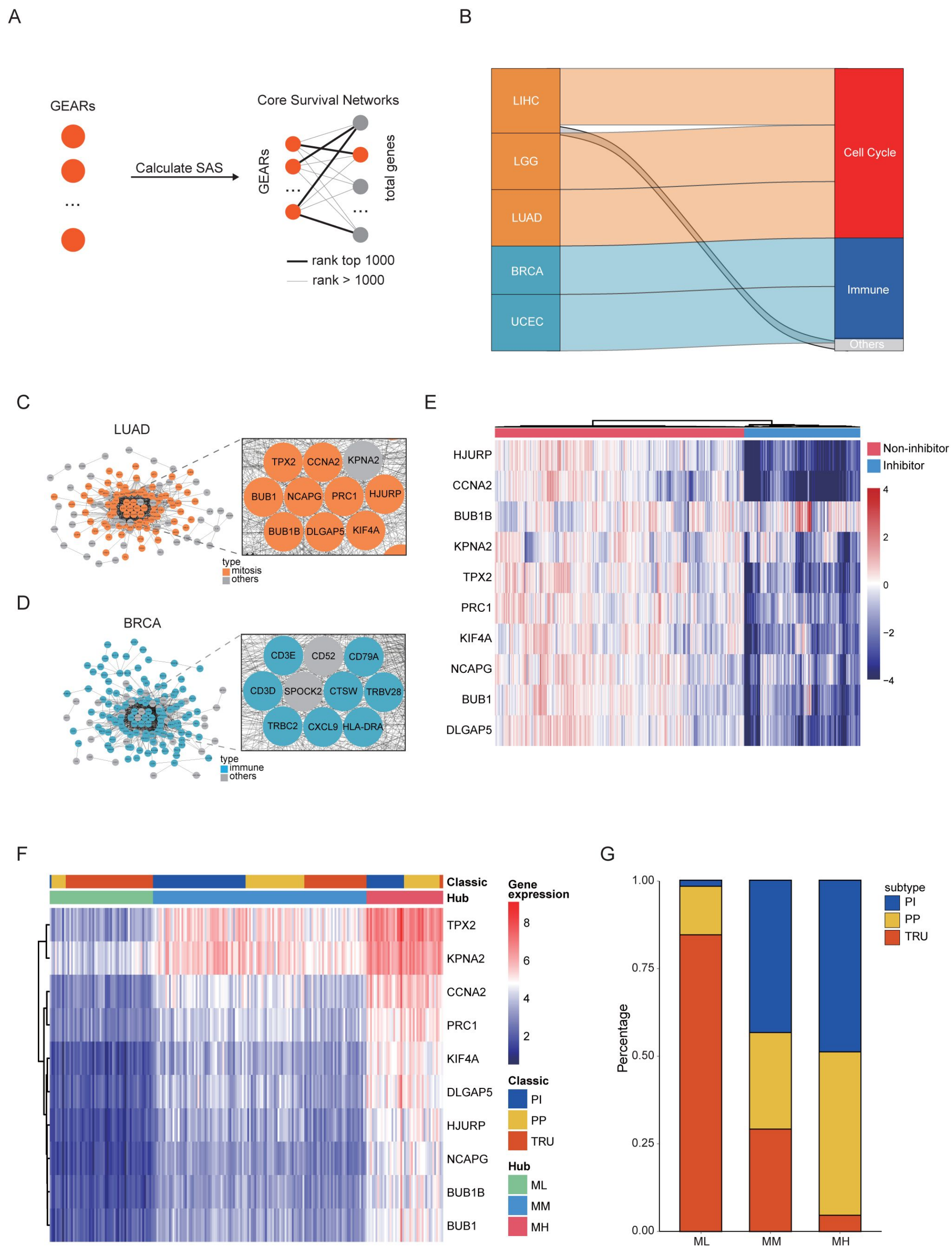


Figure 2

Identification of hub genes in mitosis or immune-related cancers.

(**A**) GEARs and core survival networks. The SAS of all GEARs was calculated, and CSN was constructed. (**B**) Cancers and their hub gene pathways. The left nodes represent five different cancers, while the right nodes detail the functional types of hub-gene pathways. The height of the edges in the middle represents the proportion of hub-gene pathways corresponding to a specified functional type. Only pathways with a Benjamini–Hochberg–adjusted $p < 0.05$ are displayed. (**C-D**) The CSN of LUAD and BRCA. The enlarged section represents the hub gene. (**E**) Heat map showing the hub gene expression of A549 after compounds treatment from cMAP database[99]. Blue annotations meant 76 compounds that can inhibit hub gene expression. (**F-G**) LUAD clustering based on hub gene expression in comparison with the classic classification.

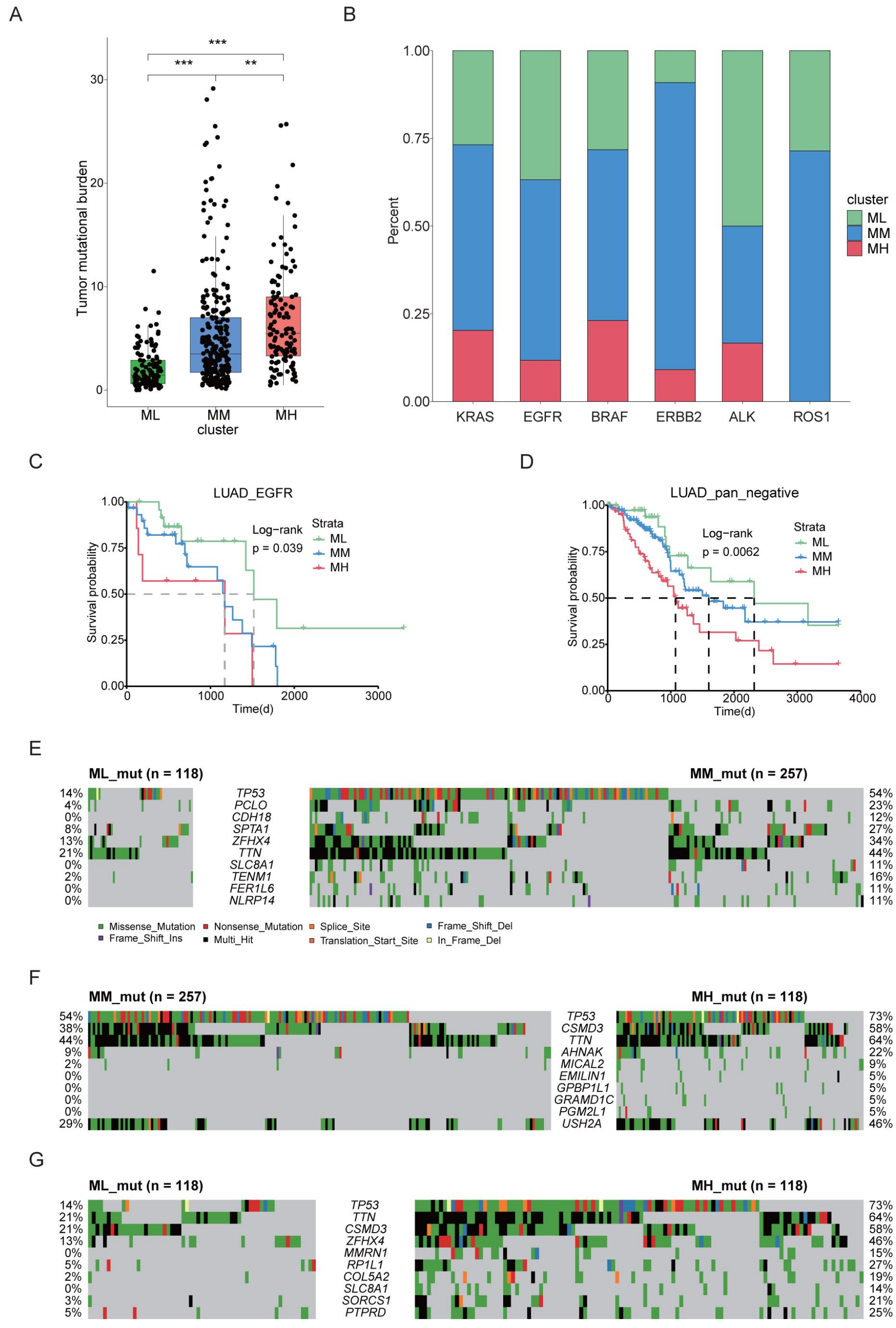


Figure 3

Different LUAD subgroups were characterized by unique genetic mutation profiles.

(**A**) TMB analysis in three groups of LUAD samples. ***P < 0.0005. (**B**) Proportion of different oncogenic drivers, including *KRAS*, *EGFR*, *BRAF* and *ERBB2* mutations, and *ALK* and *ROS1* fusions in three LUAD subgroups. (**C**) Kaplan-Meyer overall survival curves for EGFR-mutation samples of LUAD by hub gene subgroups. (**D**) Kaplan-Meyer overall survival curves for pan-negative LUAD samples by hub gene subgroups. (**E-G**) Comparison of top gene mutations in three LUAD clusters, including ML vs. MM (**C**), MM vs. MH (**D**) and ML vs MH (**E**).

We identified gene mutations that showed significant changes through gene dependence analysis (**Figure 4A**). To further explore the functional implications of these mutations, we enriched them using a pathway system called Nested Systems in Tumors (NeST)[54]. The results revealed notable differences in multiple functional pathways, including androgen receptor, cell cycle, *TP53*, *EGFR*, IL-6, and *PIK3CA*-related pathways (**Figure 4B**). To gain further insights into the functional implications of these different mutations, we assessed the gene dependency of cells with these mutations by using the DepMap database. Importantly, we observed that the pathway differences between the MM and MH clusters were primarily associated with *PIK3CA*-related pathway. Previous studies have reported that *PIK3CA* mutation confers resistance in colorectal cancer, lung cancer, and breast cancer[55–57]. Therefore, we aimed to validate the role of *PIK3CA* mutations in drug resistance by using public databases. We analyzed two lung adenocarcinoma cell lines, A549 cells and SW1573, harboring *KRAS* mutation and both *KRAS* and *PIK3CA* mutations, respectively, and identified five potentially effective inhibitors (BMS-345541, Dactinomycin, Epirubicin, Irinotecan, and Topotecan) from the previously screened set of 76 LUAD cell growth inhibitors by using GDSC data. Strikingly, SW1573 cells exhibited increased resistance to all these 5 inhibitors when compared to A549 cells (**Figure 4C**). In line with this, clinical data supported the notion that concurrent *PIK3CA* mutation is a poor prognostic factor for LUAD patients[58]. These findings suggest that *PIK3CA* mutation may contribute to drug resistance in LUAD.

Distinct clusters of BRCA exhibit different immune infiltration landscapes

In the meantime, we conducted a comprehensive analysis on BRCA, which has the largest sample size among immune-related cancers (**Figure 1C**). We hierarchically grouped the BRCA into three distinct immune subgroups based on the expression of hub genes: immune low (IL), immune medium (IM), and immune high (IH) (**Figure 5A**, Table 7). Interestingly, each PAM50 subtype included tumors from the IL, IM, and IH immune subgroups, with only minor differences in their relative frequencies[59] (**Figure 5B**, **Supplementary Figure 7A-B**). Next, we investigated the relationship among PAM50 classification, immune subtypes, and patient prognosis. Our data revealed significant prognostic differences among immune subgroups of the luminal B subtypes but not the luminal A, basal or HER2 subtypes (**Supplementary Figure 7C-F**), highlighting that integrating our method with traditional classification may enable a more detailed stratification of breast cancer samples. Mutation analysis showed that distinct patterns of gene mutations among the three subgroups had significant differences in mutation frequencies of genes such as *TP53*, *CDH1*, and *LRP2* (**Figure 5C-E**). We conducted a comparison of the proportions of immune cells across these three subgroups and observed a strong association between the overall immune cell proportion and the clustering results based on hub genes (**Figure 5F**, **Supplementary Figure 8A-J**). Specifically, the IH subgroup exhibited significantly higher proportions of $CD8^+$ T cells and Treg cells, with median percentages at 3.9% and 5.5%, respectively, compared to 1.2% and 2.8% in the IM subgroup, and 0.2% and 1.6% in the IL subgroup. Together, these findings suggest that distinct immune responses across the three subgroups potentially link genetic mutation patterns to differences in the immune microenvironment.

Next, we explored the specific genomic factors influencing immune infiltration in BRCA. Mutation analysis revealed that *CDH1* gene mutation ranked as the top two genetic alteration in BRCA samples, following *TP53* mutation (**Figure 5C-E**). Previous study has reported that *CDH1* is involved in the mechanisms regulating cell-cell adhesions, mobility, and proliferation of epithelial cells[60]. We found that *CDH1*-mutant samples exhibited significantly lower *CDH1* expression compared to *CDH1*-wildtype samples, and *CDH1* expression showed a close correlation with immune cell infiltration (**Figure 5G**, **Supplementary Figure 8K**). These results were consistent with clinical observations of *CDH1* mutation and high immune infiltration in invasive lobular carcinoma of the breast[61].

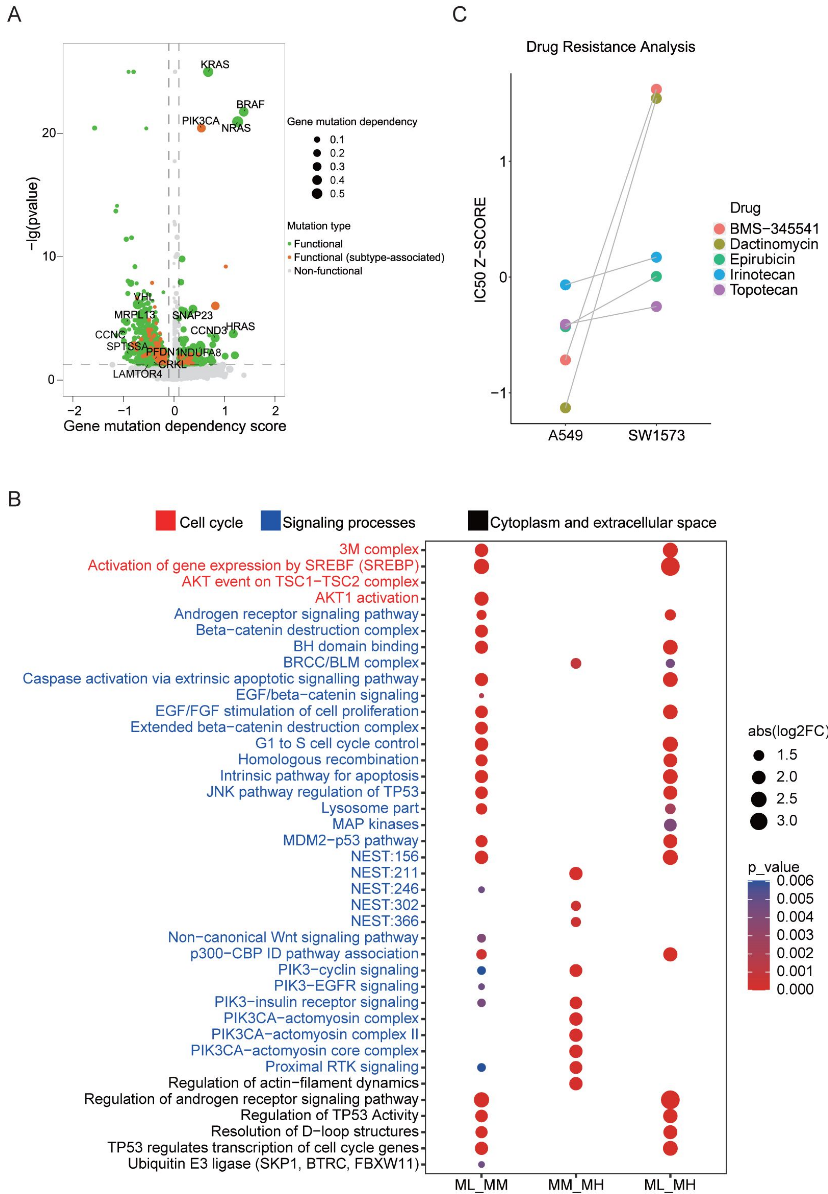


Figure 4

PIK3CA mutation associates with mitosis and drug resistance in LUAD.

(**A**) Cancer-related gene mutations were annotated based on gene dependency scores data from Depmap database. Functional mutations were indicated in green; functional (subtype-associated) mutations were highlighted in orange; non-functional mutations were indicated in gray. (**B**) The analysis of the NeST differential pathways across three groups including ML vs. MM, MM vs. MH, ML vs MH. Only pathways with a Benjamini–Hochberg–adjusted $p < 0.05$ are displayed. (**C**) Comparison of the IC50 z-score of five tumor cell growth inhibitors for A549 (*KRAS* mutation) and SW1573 (*KRAS* and *PIK3CA* mutation).

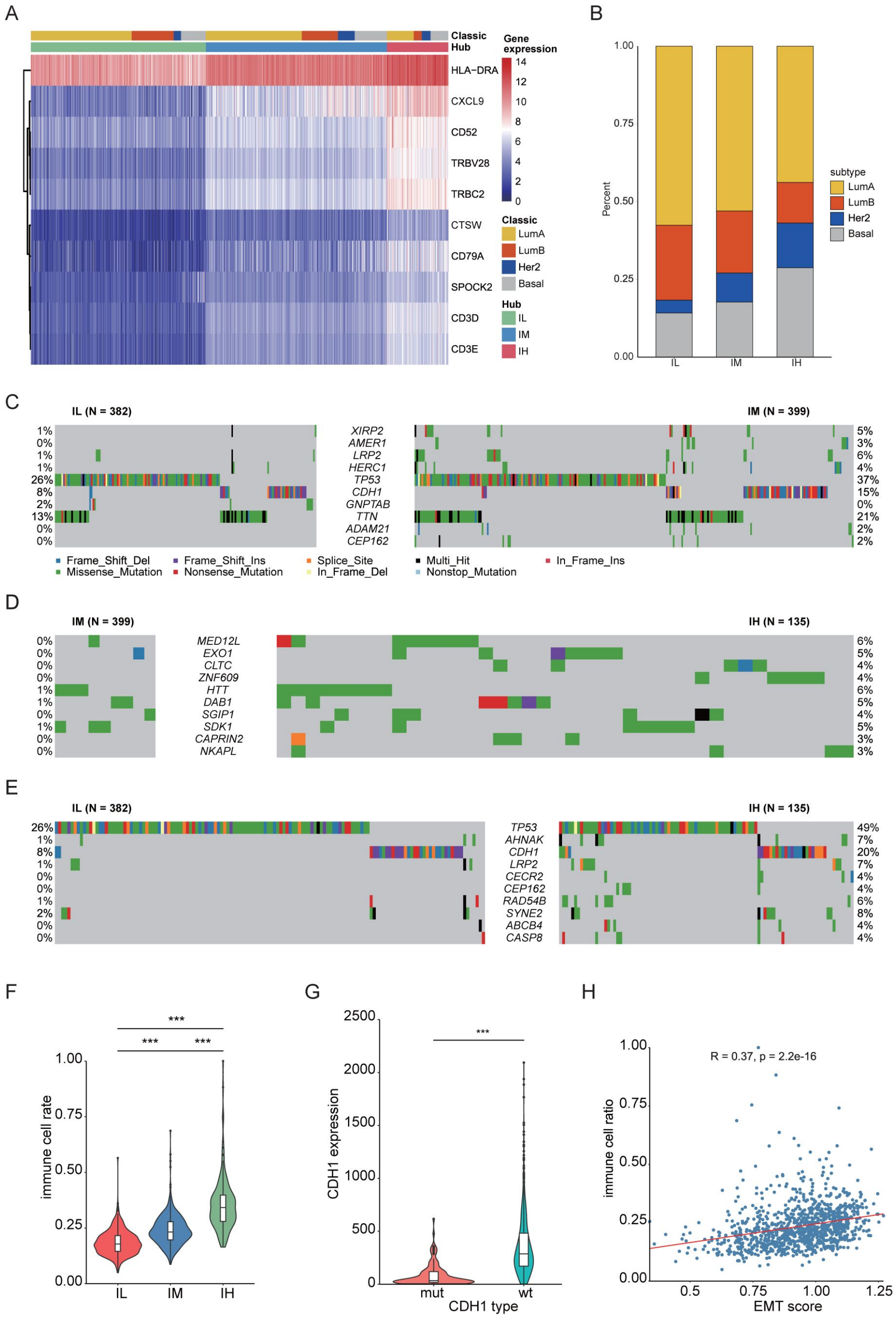


Figure 5

The hub gene classification of BRCA revealed a significant association between EMT and immune infiltration.

(**A**) Hub gene expression heatmap. The heatmap contains the result of hub gene classification and PAM50 classification of BRCA[59]. (**B**) Comparison of molecular classification based on hub genes with PAM50 classification. (**C-E**) Comparison of gene mutations in three groups of BRCA samples, including IL vs. IM (**C**), IM vs. IH (**D**), IL vs. IH (**E**). (**F**) Comparison of the total immune cell rate in three BRCA clusters. ***P < 0.0005. (**G**) Comparison of *CDH1* expression in BRCA samples with wildtype or mutant *CDH1*. ***P < 0.0005. (**H**) A schematic diagram illustrating the correlation between the EMT score and immune cell rate. The red line represents the fitting curve. The correlation analysis was performed using Pearson correlation coefficient. Statistical significance for all pairwise group comparisons was assessed with the Wilcoxon test.

The above result encouraged us to investigate how *CDH1* mutation influenced immune infiltration. In BRCA, we observed a significant correlation between the expression of *CDH1* and EMT marker genes *VIM* and *TWIST2* (**Supplementary Figure 8L-M**)[62,63]. This suggests that the regulation of *CDH1* is intricately linked with EMT process. EMT score analysis showed that there's a positive correlation between the EMT score and the proportion of immune infiltration (**Figure 5H**), which suggests that *CDH1* might influence immune infiltration through the EMT process.

Mitotic and immune signatures predict patient prognosis at pan-cancer level

Finally, we analyzed the association of *CDH1* and *PIK3CA* with specific biological processes at pan-cancer level. Our data showed that *PIK3CA* level was positively correlated with mitotic scores in most of the cancer types, whereas the correlation between *CDH1* expression and the proportion of immune cell infiltration was positive in PRAD, LGG, OV, Uveal Melanoma (UVM), THCA, LIHC and LUAD but negative in BRCA, Testicular Germ Cell Tumors (TGCT), Thymoma (THYM), LUSC, BLCA, Head and Neck squamous cell carcinoma (HNSC), Sarcoma (SARC), Esophageal carcinoma (ESCA), PAAD and STAD (**Figure 6A-B**). Eventually, we sought to explore the prognostic relevance of mitosis and immune to patient outcomes at the pan-cancer level. The scores for these biological processes were computed using RNA-seq data from the TCGA database, and the median score was used as a cut-off to categorize patients into different subgroups. Out of 33 cancer types, 19 showed a statistically significant correlation ($p < 0.05$) with at least one pathway score (**Figure 6C**). Specifically, 10 cancer types were exclusively associated with the mitosis score, 4 cancer types were exclusively associated with the immune score, and 5 cancer types exhibited a correlation with both mitosis and immune scores simultaneously. Overall, the identification of mitosis and immune-related biological processes as significant prognostic factors at the pan-cancer level suggests their potential utility as valuable biomarkers for predicting patient prognosis.

Discussion

Due to the multifaceted nature of variables affecting cancer patient prognosis, it remains uncertain whether there exist a set of genes steadily associated with cancer prognosis, regardless of sample size and other factors. Here, we utilized the MEMORY method to address this question and discovered that all the cancer types have GEARs. We observed significant variation in the number of GEARs among different cancer types, indicative of cancer type-specific patterns. The substantial heterogeneity in driver genes and mortality rates among various cancer types could potentially explain this phenomenon. For example, THCA, known for its low malignancy, displays the fewest genetic expression alterations among all the studied cancer types. This observation could be correlated with the relatively high five-year survival rate in THCA, which exceeds 50% even in advanced stage[64]. In contrast, PRAD, despite of a favorable prognosis similar to THCA, exhibits a significantly higher number of genetic expression alterations. We hypothesize that the positive prognosis in PRAD cases might be largely attributed to early diagnosis, which enables timely treatment for the majority of PRAD patients[65]. This discrepancy between the number of genetic alterations and prognosis in THCA and PRAD highlights the intricate nature of cancer genetics and emphasizes the importance of personalized considerations in cancer treatment and prognosis evaluation. Furthermore, certain genes are commonly found in the GEARs of various cancer types, indicating their potential significance in cancer development. For instance, *BEX4* is present in the GEARs of LUAD, KIRC, LGG, PAAD, and STAD, and known to play oncogenic role in inducing carcinogenic aneuploid transformation via modulating the acetylation of α-tubulin[66,67]. Aneuploidy is a hallmark characteristic of cancer and can lead to alterations in the dosage of oncogenes or tumor suppressor genes, thereby influencing tumor initiation and progression[68–71]. The regulation of aneuploid transformation by *BEX4* may represent a common mechanism through which this gene impacts prognosis across different cancer types.

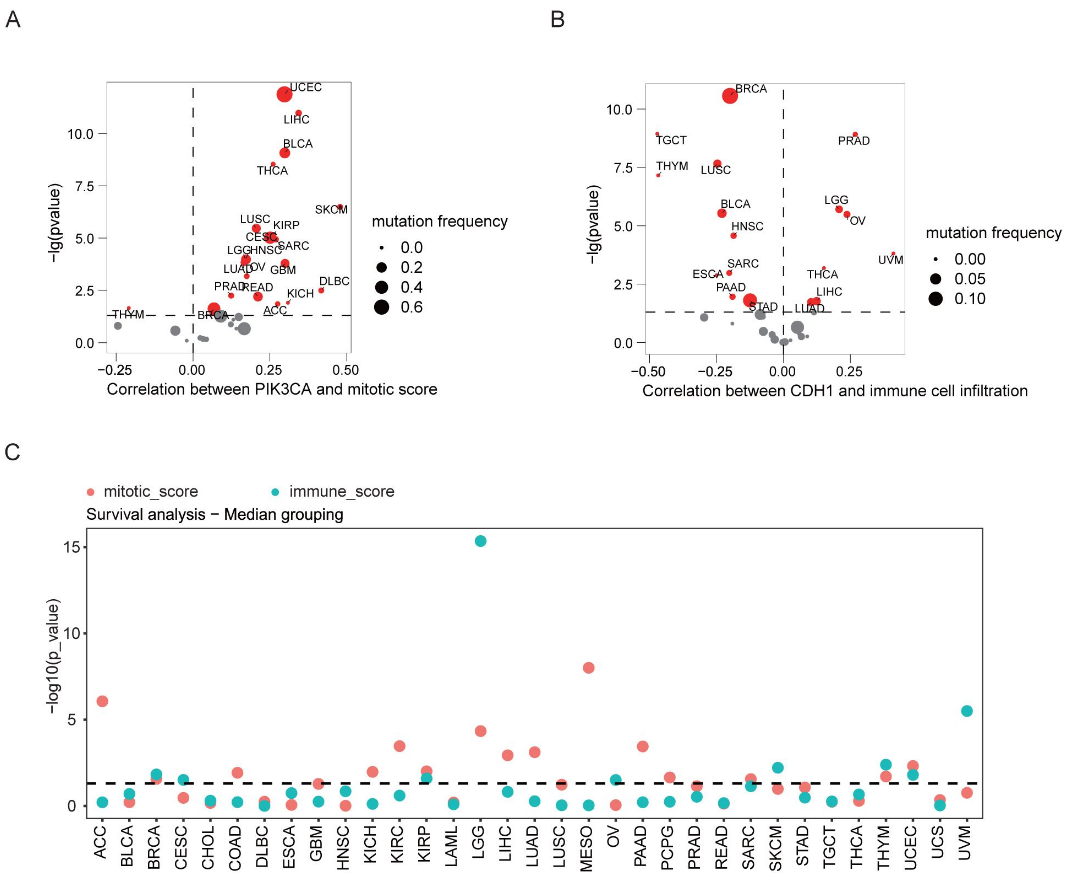


Figure 6

Mitosis and immune signatures predict patient prognosis at the pan-cancer level.

(**A**) The correlation of mitosis scores of 33 TCGA cancer types with *PIK3CA* expression. (**B**) The correlation of immune cell infiltration rate of 33 TCGA cancer types with *CDH1* expression. (**C**) The mitosis and immune-related pathway scores of 33 cancer types. Median score was utilized as a threshold to categorize patients for survival analysis. Among the 33 cancer types examined, 10 cancer types were exclusively associated with the mitosis score, 2 cancer types were exclusively associated with the immune score, and 5 cancer types showed a correlation with both mitosis and immunity scores concurrently. Dots represent -=lo$_{10}$=(O values from log-rank tests comparing high vs. low score groups for each cancer type. The dashed line marks the nominal significance threshold (p = 0.05).

Subsequently, we discovered that GEARs across different cancers displayed distinct functional characteristics. Notably, a recurrent theme was the prominence of mitosis-related and immune-related features. Specifically, the GEARs of LGG, LIHC, and LUAD were enriched in mitosis-related pathways, whereas BRCA and UCEC showed the enrichment of immune-related pathways. Mitosis and immune processes have been widely recognized as having a significant impact on patient prognosis[72–76]. However, current clinical guidelines do not yet recommend the use of transcriptomic data to assess scores related to mitotic and immune pathways for predicting patient outcomes, despite the well-established association between these pathways and prognosis in cancer[77–82]. Cancers enriched with mitosis pathways often exhibit heterogeneous tumor growth kinetics across individuals, with tumor size being one of the crucial factors influencing patient survival[83–85]. For instance, in the case of LGG, the growth rate of tumors directly impacts the patient's prognosis due to the primary location in the brain. Consequently, alterations in the expression levels of genes related to mitosis are often indicative of the prognosis of LGG[86]. Cancer types that are enriched in immune-related pathways, such as BRCA and UCEC, are closely associated with deficiencies in DNA mismatch repair (MMR)[87,88]. Cancers enriched in immune-related pathways often exhibit heterogeneous tumor growth kinetics across individuals, where tumor size is closely correlated with patient survival.

To explore the underlying differences between samples associated with distinct mitotic and immune-related pathways, we employed GEAR analysis to identify hub genes for further molecular classification and mutation analysis. Specifically, we focused on LUAD and BRCA, two representative cancer types exhibiting enriched mitosis and immune signatures in GEARs. By classifying the hub gene, we divided LUAD into three subgroups: ML, MM and MH, which displayed significant difference in survival outcomes. Subsequently, we utilized NeST to analyze the mutations within these subgroups. Notably, there were significant differences observed in cell cycle and signal transduction-related pathways among ML, MM and MH subgroups. Of particular importance, the *PIK3CA*-related pathway emerged as a key differentiating pathway between the MH and MM subgroups. Our analyses suggest that LUAD cell line carrying *PIK3CA* mutation may exhibit increased drug resistance. This aligns with previous studies demonstrating that targeting the PI3K pathway can overcome drug resistance[89,90]. Of course, future research is warranted to elucidate the exact functional role of *PIK3CA* mutations in LUAD.

To investigate the factors influencing immune infiltration in BRCA, we classified the samples based on hub genes and analyzed the differentially mutated genes. Interestingly, we found that *CDH1* mutation occurred at a high frequency in the IH subgroups. This led us to speculate that *CDH1* plays a crucial role in the regulation of immune infiltration in BRCA. Furthermore, previous studies have reported that *CDH1* inactivation promotes immune infiltration in breast cancer[61]. *CDH1* is a vital gene associated with EMT, and various studies have demonstrated the close relationship between EMT and the tumor immune microenvironment in different cancers[91]. Consistently, our results also supported the correlation between EMT score and immune infiltration in BRCA. These findings suggest that the mutations in *PIK3CA* and *CDH1*, identified through GEAR analysis, have significant impacts on cancer development and hold potential value in improving clinical therapies. This further emphasizes the importance of GEARs in understanding cancer biology and guiding treatment strategies.

Lastly, we investigated the prognostic predictive capabilities of the mitosis score and immune score at the pan-cancer level. Surprisingly, we found that approximately half of the cancer types exhibited significant correlations between these two scores and patient prognosis. Interestingly, even for cancers originating from the same primary location, their correlations with these scores could differ, indicating the potential diversity of mechanisms underlying cancer-related mortality. For instance, both LGG and GBM are brain tumors, but the primary risk factor for patient prognosis in LGG is closely linked to tumor diameter, whereas the main risk factor for GBM patients lies in its high invasiveness and challenge of surgical resection[92,93].

Although the present study defined a preliminary catalogue of GEARs and supported their relevance with extensive biological data, we recognize several outstanding methodological limitations at both the algorithmic and analytical levels. Chief among them is the need for rigorous, large-scale multiple-testing adjustment before any GEAR list can be considered clinically actionable. Because this work was conceived as an exploratory screen, such correction was intentionally deferred; nonetheless, forthcoming versions of MEMORY will incorporate a dedicated false-positive–control module that will be applied to the consolidated GEAR catalogue prior to any translational use. Although GEARs show robust prognostic associations, the CSN edges are undirected, and hub genes serve primarily as stable biomarkers. Future work will explore causality and therapeutic implications. As a result, the hub genes identified from GEAR in the CSN may primarily serve as stable and effective biological markers. Additionally, through multi-omics analysis, we obtained some functional mutations, but the therapeutic significance of these mutations remains to be elucidated. For example, further study is needed to understand the exact role of *PIK3CA* mutations in promoting tumor cell proliferation and drug resistance. Similarly, the association of *CDH1* mutations with the infiltration of multiple immune cell types also warrants additional experimental investigation. In our future studies, we plan to utilize protein-protein interaction networks and pathway databases to construct a novel network based on the CSN. This approach will allow us to directly screen genes from GEARs that could potentially serve as therapeutic targets. Undoubtedly, future efforts are still required to utilize protein-protein interaction networks and pathway databases to construct a new network based on CSN.

In conclusion, our study utilized the MEMORY algorithm to identify GEARs in 15 cancer types, highlighting the significance of mitosis and immunity in cancer prognosis. Our findings demonstrate that GEARs possess substantial biological significance beyond their role as prognostic biomarkers. This study provides valuable guidance for establishing standards for survival analysis evaluation and holds potential for the development of novel therapeutic strategies.

Methods

Datasets

The gene expression profiles and clinical information of 33 cancers were obtained from the TCGA database and downloaded by the GDC data website (*https://portal.gdc.cancer.gov/*). All gene expression data and survival data were integrated and normalized.

Multi-gradient permutation survival analysis

RNA-seq count data were normalized by TPM form the gene expression matrix (genes × samples) for each cancer type. We randomly sampled the gene expression data according to the gradient. The sampling strategy was as follows: Ten gradient increases in sample size were pre-set, ranging from approximately 10% to about 100%, with intervals of 10%. Random samples were taken from total samples of each cancer 1000 times, based on each pre-set sample size. Multiple sampling matrix was obtained after the sampling strategy was performed in each gradient. Then survival analysis was performed using the R package “survival”. Gene expression values were dichotomized based on the median expression level of each gene in the sampled matrix. The survival analysis method was as follows according to the median expression value of every gene, every sampling matrix was divided into a high and low expression group. This approach divides the samples into high and low expression groups with equal size, providing a standardized grouping strategy for survival analysis. We used 1 for significant survival analysis results ($P < 0.05$) and 0 for non-significant results ($P > 0.05$). The survival analysis matrices were obtained after these processes. This generated a binary matrix of shape (genes × 1000) per gradient, representing the significance profile across permutations. The binary matrix was integrated to a significant-probability matrix by a formula:

$$A_{ij} = \frac{\sum_{n=1}^{1000} k_{j \times i \times n}}{1000} \qquad (1)$$

where the A_{ij} is the value from row i and column j in the significant-probability matrix. We defined the sampling size k_j reached saturation when the max value of column j was equal to 1 in a significant-probability matrix. The least value of k_j was selected, and the genes with their corresponding A_{ij}greater than 0.8 were extracted as GEAR.

Construction of core survival network

We also defined survival analysis similarity (*SAS*) as the similarity of the effect on patient prognosis in two genes. For each gene, the 1 000 permutation tests performed at the first sampling gradient yield a binary vector $A = (A_1, \ldots, A_{1000})$, $B = (B_1, \ldots, B_{1000})$ (1 = significant, 0 = non-significant). Vectors from the first sampling gradient (kD, approximately 10% of the cohort) were used as input. At this sample size, significance patterns exhibited high variability, with many genes not yet consistently significant. This variability was utilized to enable SAS to differentiate gene pairs with concordant versus discordant behavior.

$$\text{Let } a = \sum_{i=1}^{1000} A_i,\ b = \sum_{i=1}^{1000} B_i,\ c = \sum_{i=1}^{1000} A_i B_i$$

where c is the number of permutations in which both genes are significant and a, b are the total significant counts for each gene.

SAS is calculated as

$$SAS(A, B) = \frac{c}{a + b - 2c + 1} \qquad (2)$$

a Jaccard-like metric bounded between 0 (no overlap) and 1 (complete overlap); the “+1” term prevents division by zero.

Pair-wise SAS values were computed between every GEAR and all other genes. The 1 000 SAS were selected to construct a core survival network (CSN) in Cytoscape[94]. Node degrees were calculated, and the top 10 GEARs with the highest degree in these networks, selected from those with mean expression >10 TPM, were defined as hub genes. We constructed the Core Survival Network (CSN) using the top 1000 gene pairs ranked by SAS values. As the CSN was designed as a heuristic similarity network to prioritize genes for exploration, edge selection was based on empirical ranking rather than formal statistical thresholds.

Gene ontology enrichment analysis

Gene ontology has been used to classify genes based on functions. The gene functions were divided three types, includes molecular function (MF), biological process (BP), and cellular component (CC). ClusterProfiler is an R package for gene set enrichment analysis[95]. The gene ontology (GO) enrichment analysis was processed in GEARs and hub genes by ClusterProfiler.

Hub gene classification

Tumor samples were genotyped using RNA-seq data. The following steps outline the genotyping and clustering methods used in this study. The expression matrix of hub genes corresponding to various cancer types was extracted from the RNA-seq data. This involved filtering the RNA-seq data to retain only the expression levels of the identified hub genes. To normalize the expression data and reduce the impact of extreme values, a pseudo count of 1 was added to each expression value, and the resulting matrix M was log-transformed using the formula: $M' = log_2 (M + 1)$. The log-transformed matrix M' was then subjected to hierarchical clustering. This clustering was performed in R (v4.3) using the Ward's method (hclust, method = "ward.D2") to minimize the variance within clusters. The distance metric used was the Euclidean distance (dist, method = "euclidean"). The number of clusters was set to 3 using the cutree function (cutree(model, k = 3)), based on visual inspection of the dendrogram structure.

Calculation of tumor mutation burden (TMB) and differential mutation

The TMB and differential mutation gene analysis were carried out to explore the differences between different genotyped samples at the genomic level. The data are from the gene mutation data of TCGA tumor samples, and the grouping information is from the hub gene hierarchical clustering results. Maftools is an R package for the analysis of somatic variant data, which can export results in form of charts and graph[96]. The calculation of TMB and difference mutation analysis were used by the maftools.

Quantifying the effect of gene mutations for tumor cell viability

Gene dependence refers to the extent to which genes are essential for cell proliferation and survival. The Cancer Dependency Map (Depmap, *https://depmap.org/portal/*) database provides genome-wide gene dependence data for large number of tumor cell lines[97]. Gene mutation data of cell lines were obtained from the CCLE database (*https://sites.broadinstitute.org/ccle*). The genes appearing in the gene mutation data of cell lines were extracted and sorted into mutation list. Each gene in the list was then analyzed for survival-dependent differences. For each gene in the mutation list, we compared the gene dependence score (S) between cell lines with and without the gene mutation. Specifically, cell lines from the CCLE mutation dataset harboring a mutation in a given gene were classified into one group, while cell lines without the mutation constituted the control group. S for the mutation (S_m) and wild-type (S_{wt}) cell lines were then extracted from the DepMap database.

A two-sample t-test was used to assess the statistical significance of differences between S_m and S_{wt}. The test yielded a P-value, with an alpha level set at 0.05, to determine the significance of the difference in means. D_m was defined as gene mutation dependence, the computational formula was as follows:

$$D_m = |\bar{s}_m - \bar{s}_{wt}| \qquad (3)$$

Subsequently, we conducted a standardization assessment of D_m, employing the following formula:

$$S_{dm} = \frac{2(\bar{s}_m - \bar{s}_{wt})}{\bar{s}_m + \bar{s}_{wt}} \qquad (4)$$

In this study, we categorized mutations identified in cell lines into three groups based on their impact on gene dependency. Functional mutations refer to the mutations that exhibit significant changes in gene dependency scores, with a P-value < 0.05 and $S_{dm} > 0.1$. Non-functional mutations were those without significant changes in gene dependency scores compared to wildtype. Subtype-associated functional mutations were a subset of functional mutations that show statistically significant differences in occurrence frequency across different LUAD subtypes.

Identification of drugs downregulating hub gene expression

The IC50 data of 76 compounds of LUAD cell lines were obtained from Genomics of Drug Sensitivity in Cancer (GDSC) database[98]. The hub gene expression data of A549 cell lines after treatment with 76 compounds was obtained from Connectivity MAP (cMAP, *https://clue.io/*)

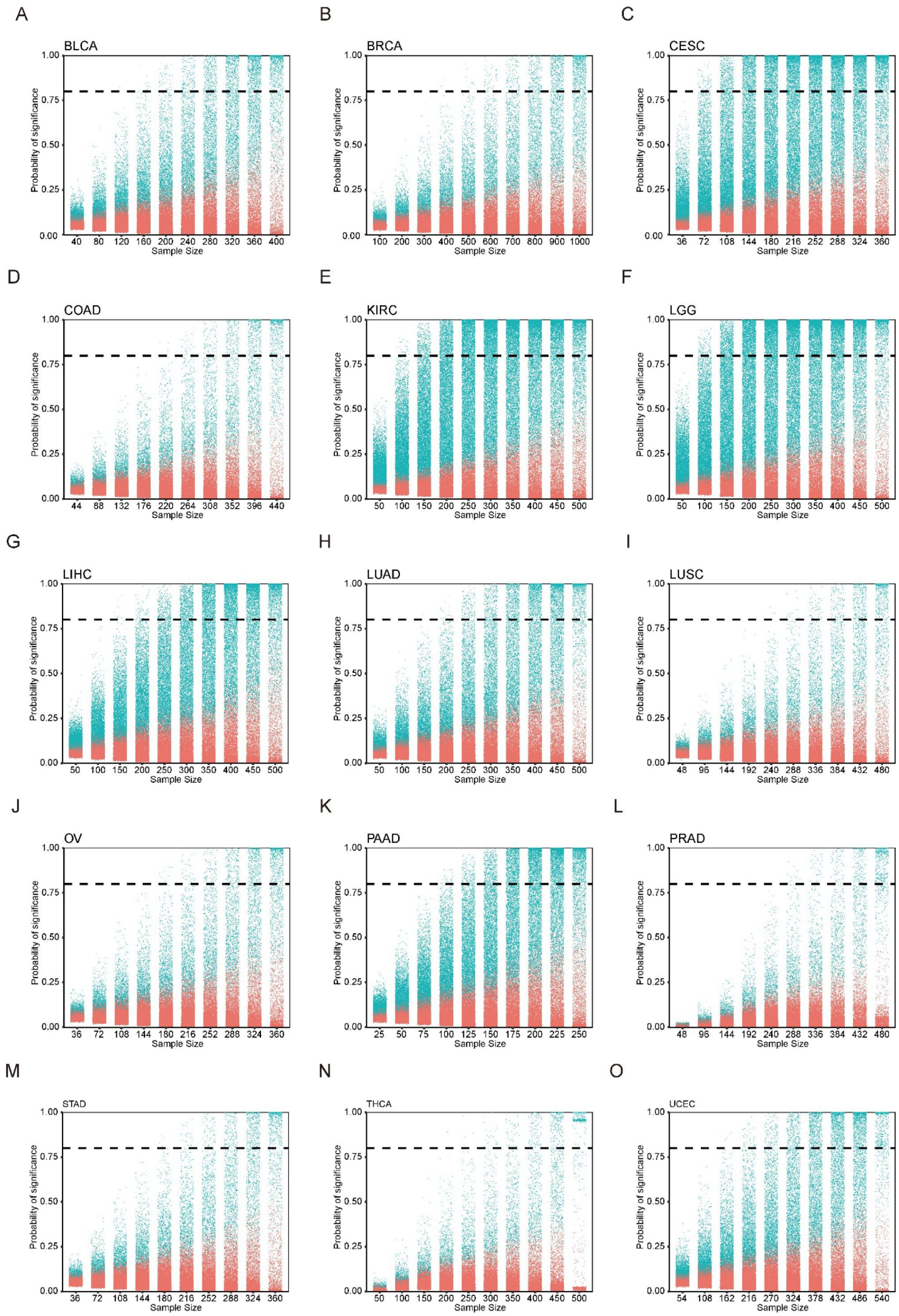
database, and the data were grouped by hierarchical clustering[99]. The drugs which were presented only in the hub gene expression suppression group were considered to effective against the LUAD cell.

Immune infiltration analysis

Immune infiltration data was calculated by quantiseq method[100]. Immunedeconv was an R package for unified access to computational methods for estimating immune Cell fractions from bulk RNA-seq data[101]. The data were from the gene expression data of TCGA tumor samples after TPM was standardized. Then the immune cell fractions of tumor samples were calculated by quantiseq method invoking immunedeconv.

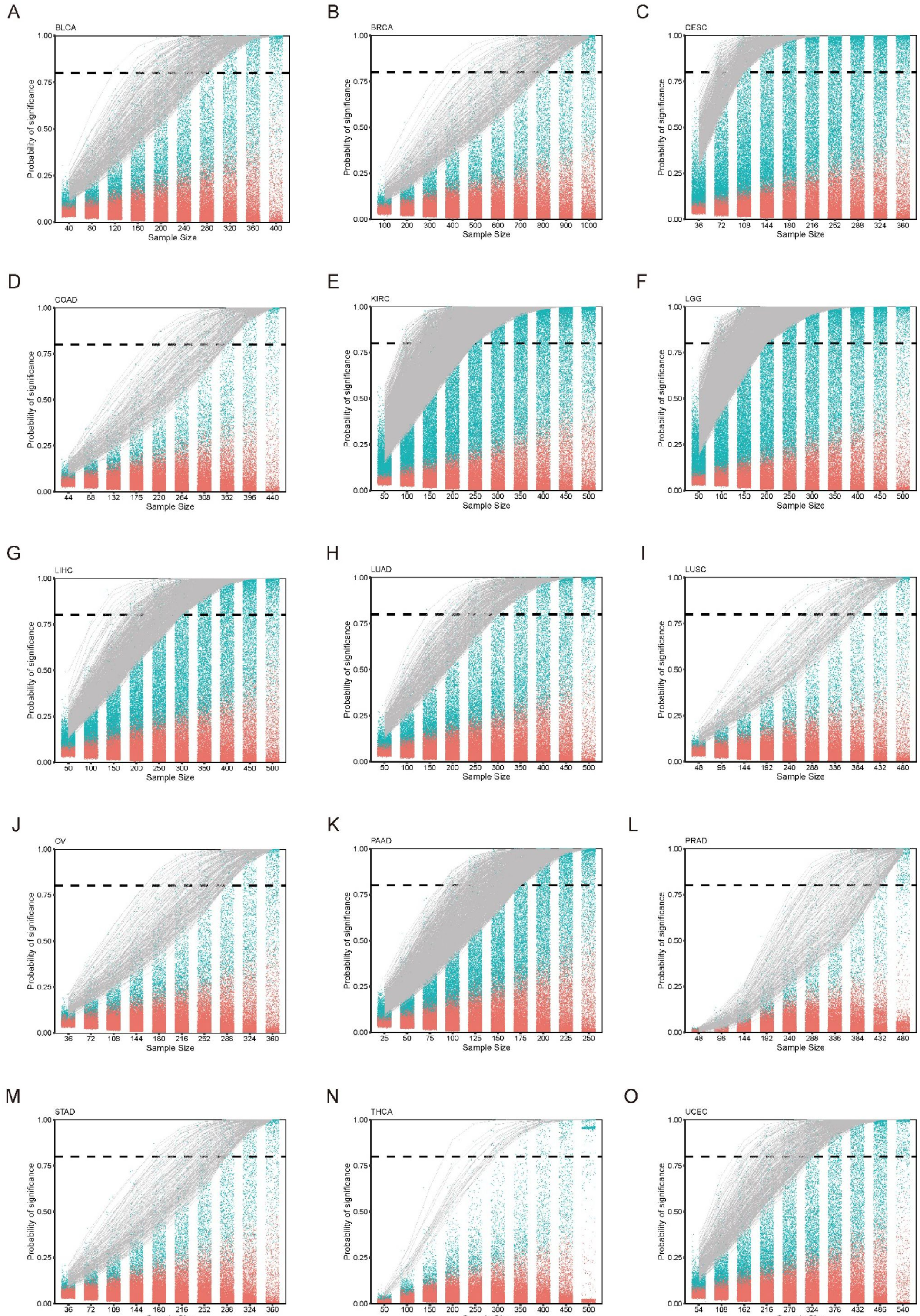
Biological function score calculation

GSVA is an R package for calculate biological function score based on a single sample[102]. The data was from TCGA in the calculation process, and the software package was GSVA. The immune score was T-cell infiltration rate of that calculated by quantiseq[100]. The gene sets that calculated the mitosis score were obtained from gene ontology (GO:0140014).



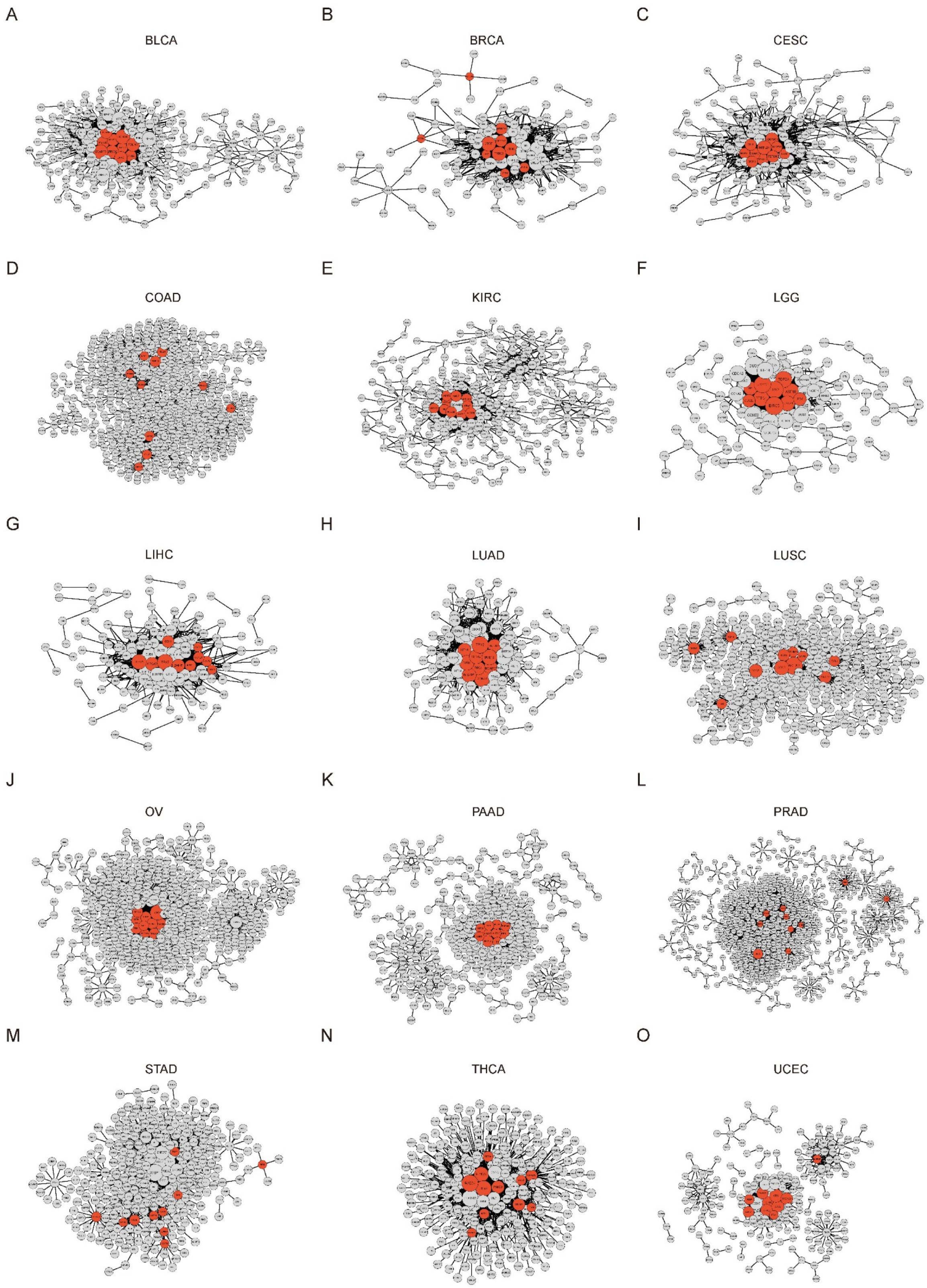
Supplementary Figure 1

(A) to (O) The number of genes significantly related to prognosis which were obtained through MEMORY was positively correlated with sample size. The vertical value of the dotted line is 0.8.



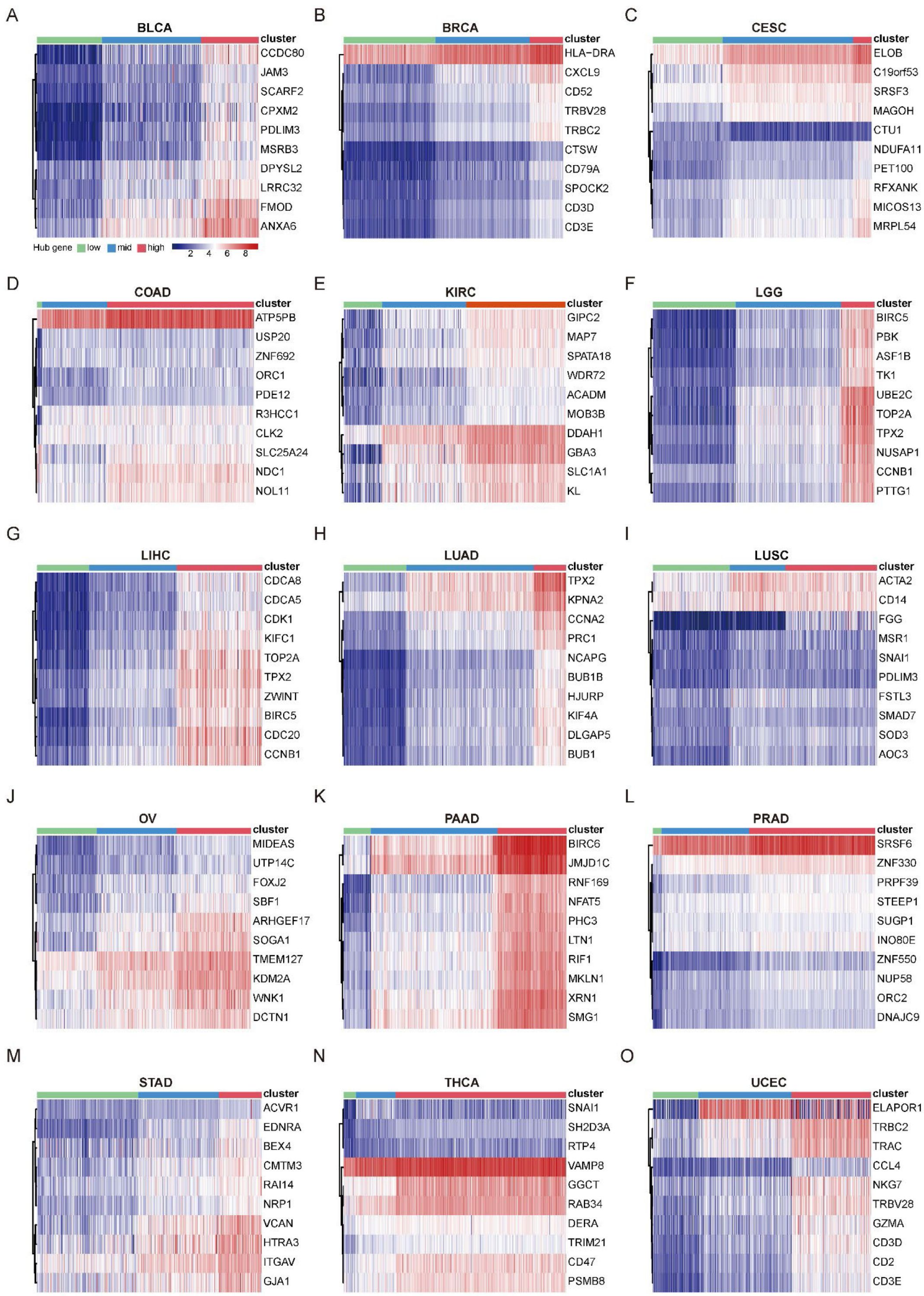
Supplementary Figure 2

(A) to (O) The significant probability of GEAR at each gradient. The vertical value of the dotted line is 0.8.



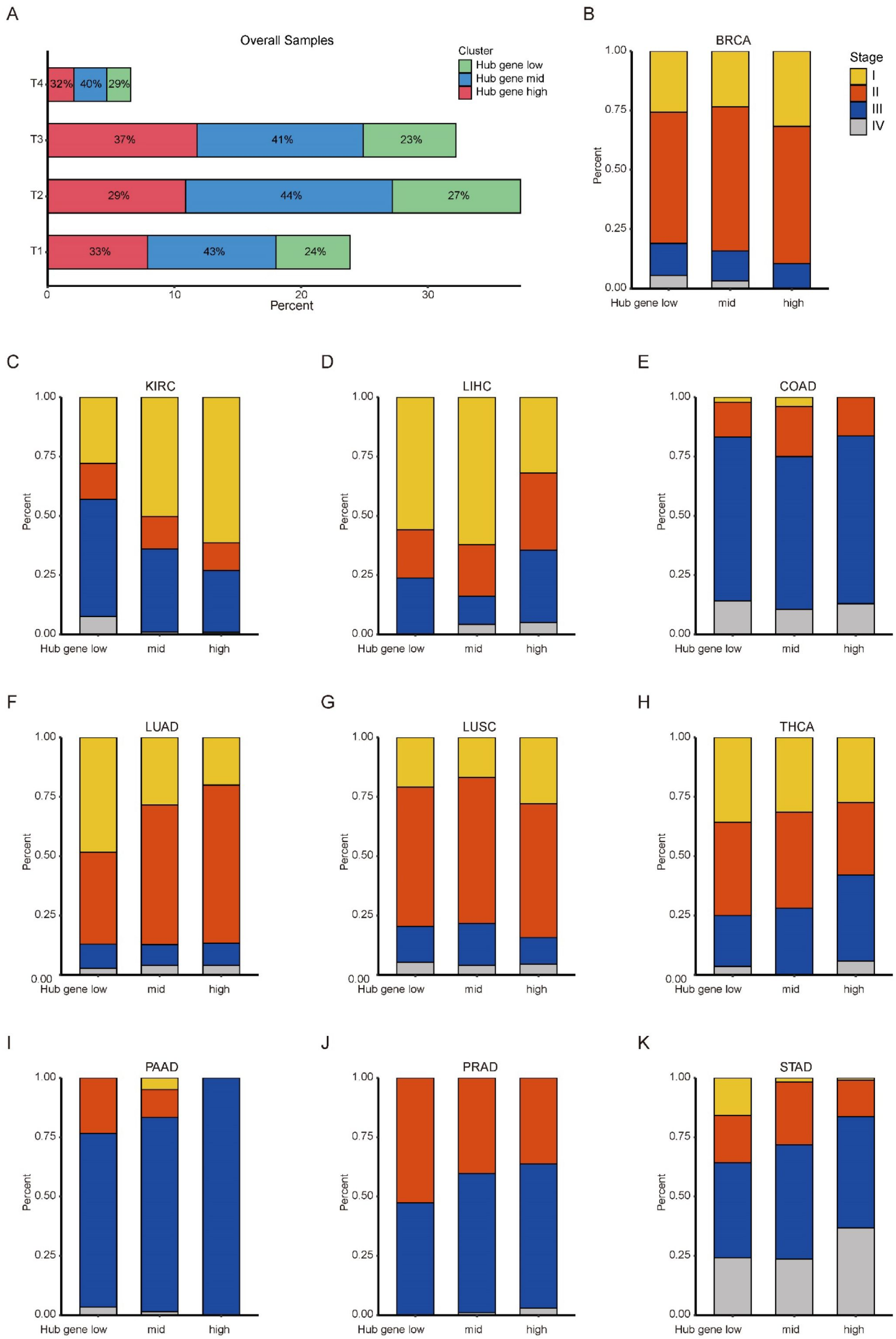
Supplementary Figure 3

(A) to (O) The core survival networks of 15 cancer types. Red nodes are hub genes.



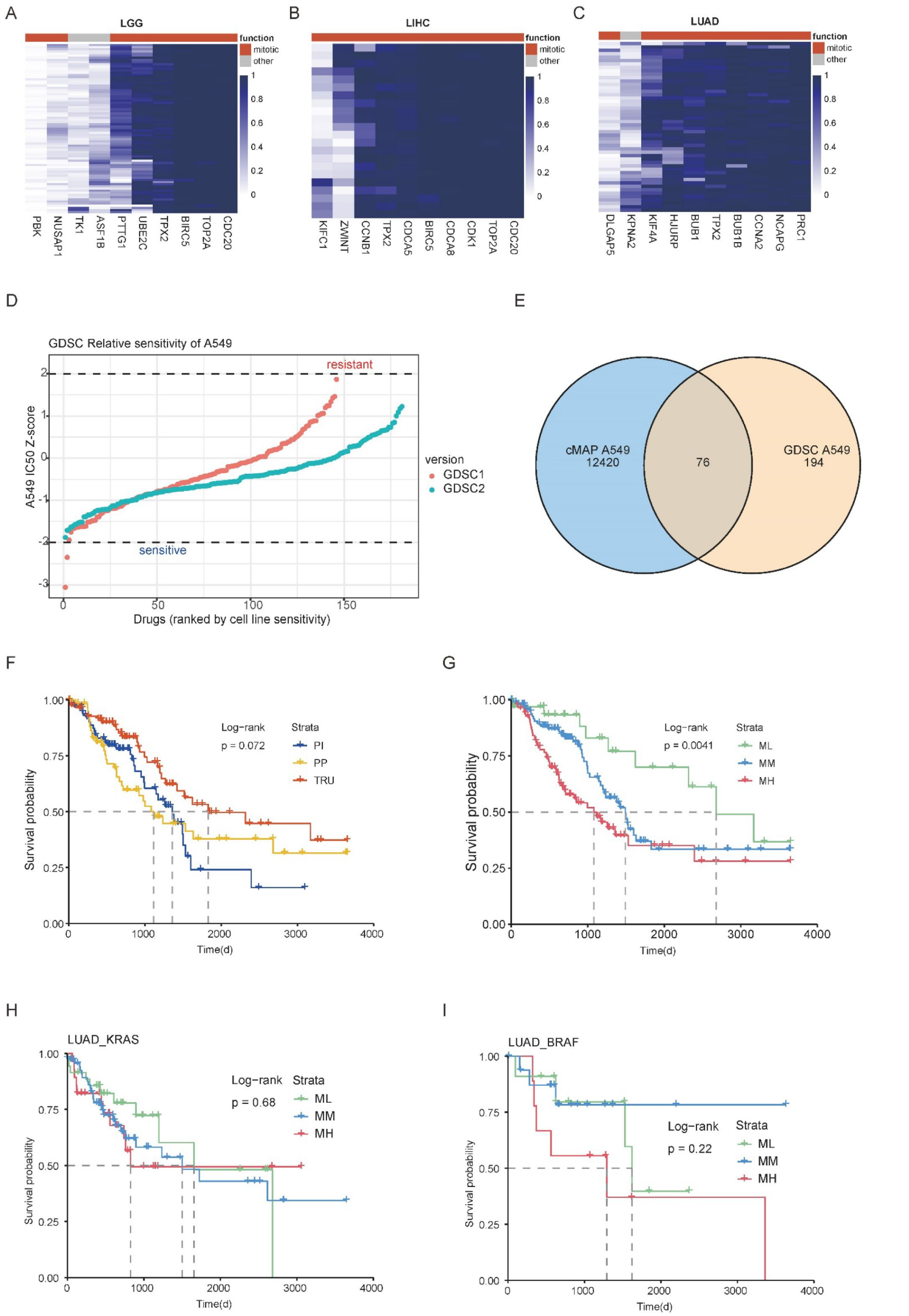
Supplementary Figure 4

(A) to (O) 15 cancer types were genotyped through hierarchical clustering based on the expression of hub genes, and the samples of each cancer were divided into three clusters



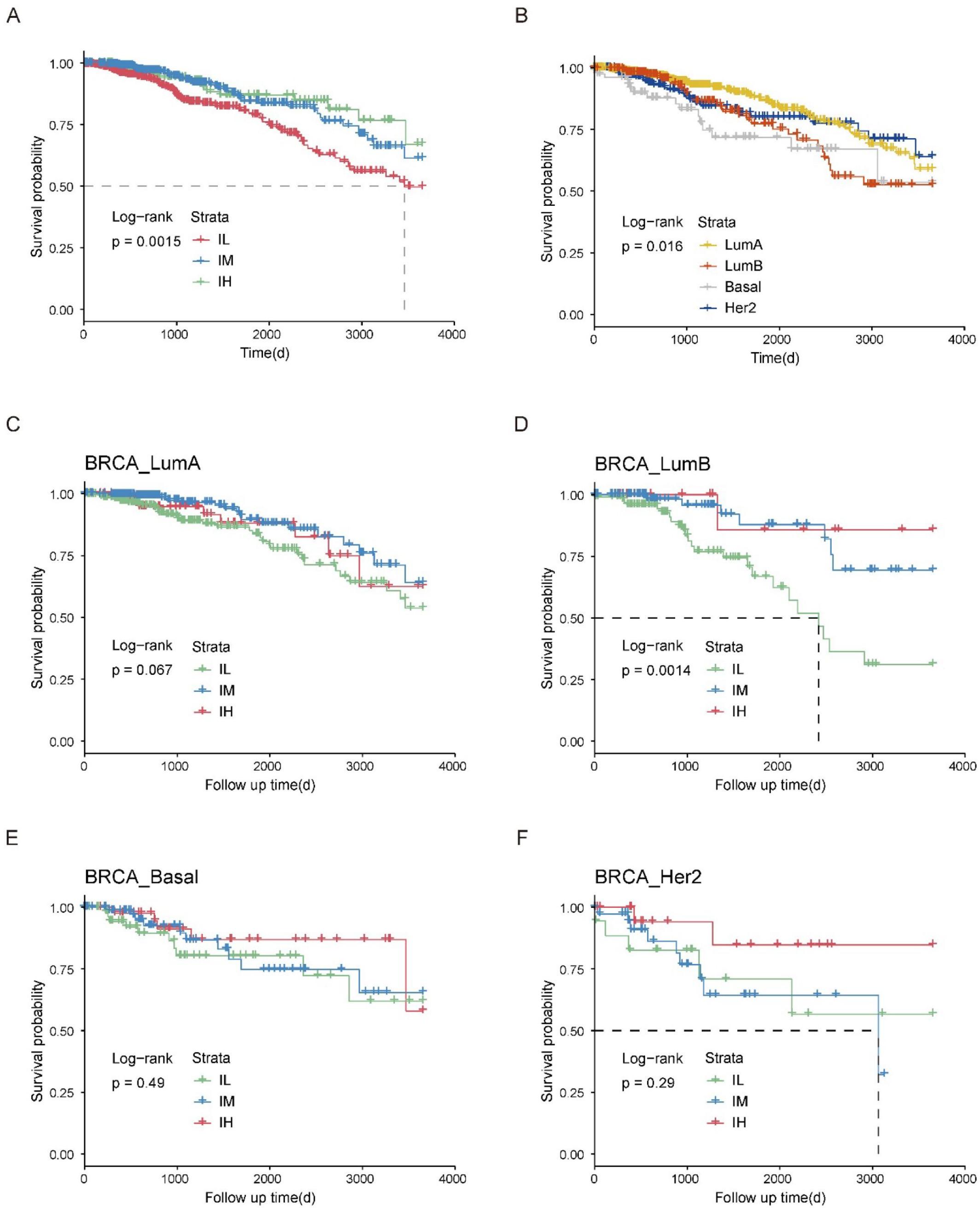
Supplementary Figure 5

(A) The proportion of hub gene classifications contained in different stages of pan-cancer level. (B) to (K) The proportion of cancers at different stages contained in different hub gene classifications.



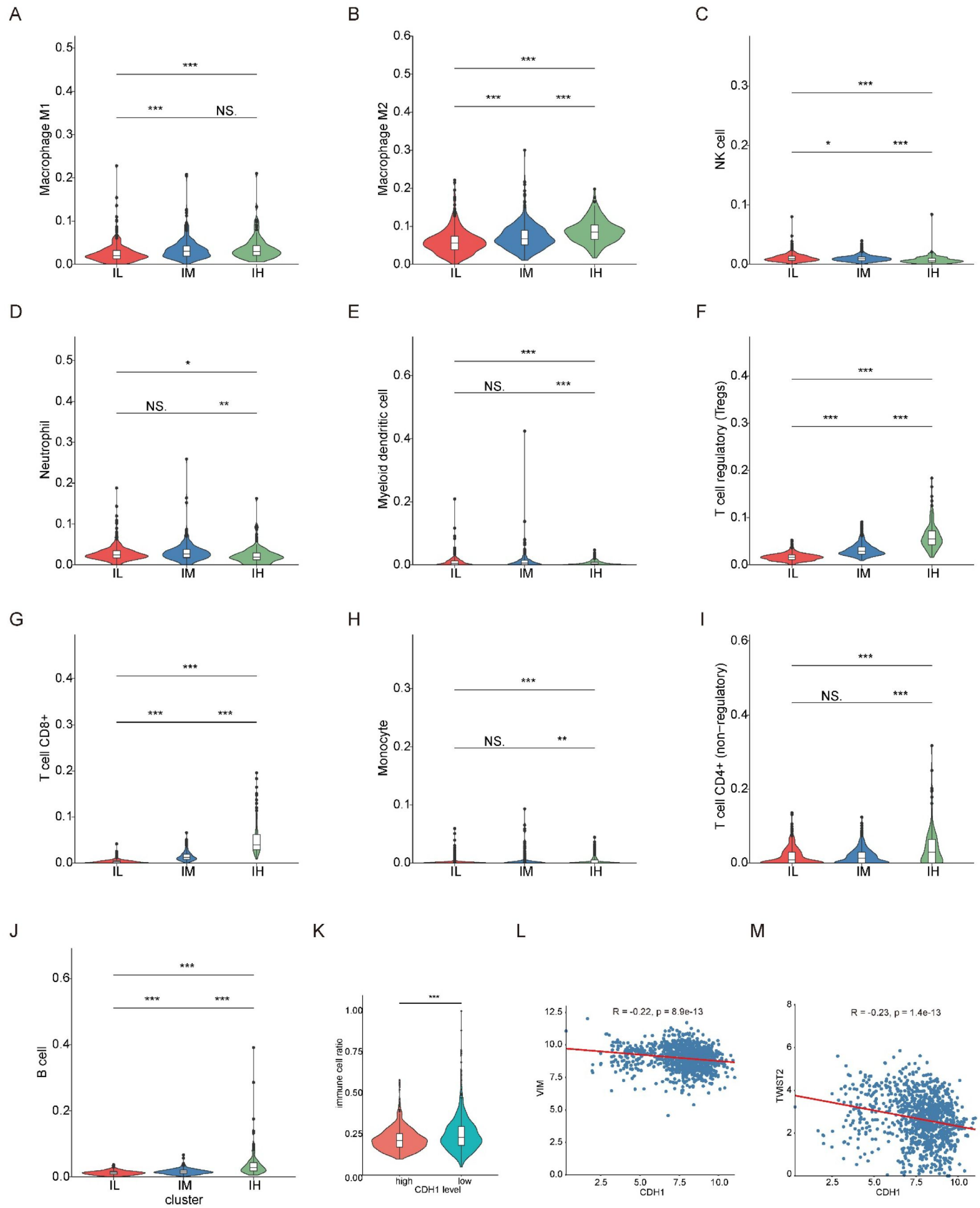
Supplementary Figure 6

(A) to (C) The gene dependency analysis of hub genes for LGG (A), LIHC (B), LUAD (C). (D) The number of compounds associated with A549 in the cMAP and GDSC databases. (E) Relative sensitivity of A549 cell line to all compounds from the GDSC database, calculated based on IC50 values. (F) Kaplan-Meyer overall survival curves for LUAD classic subgroups. (G) Kaplan-Meyer overall survival curves for LUAD hub gene subgroups. (H) Kaplan-Meyer overall survival curves for *KRAS*-mutation samples of LUAD by hub gene subgroups. (I) Kaplan-Meyer overall survival curves for *BRAF*-mutation samples of LUAD by hub gene subgroups.



Supplementary Figure 7

(A) Kaplan-Meyer overall survival curves for BRCA classic subgroups. (B) Kaplan-Meyer overall survival curves for BRCA PAM50 subgroups. (C-F) Kaplan-Meyer overall survival curves for BRCA samples by hub gene subgroups in LumA (C), LumB (D), Basal (E), Her2 (F).



Supplementary Figure 8

(A) to (J) Comparison of infiltration of different types of immune cells in three group BRCA samples. ***P < 0.0005, **P < 0.005 and *P < 0.05. (K) Comparison of the proportion of immune cells in the total cells in BRCA samples with high *CDH1* expression and low *CDH1* expression. ***P < 0.0005. (L) Schematic diagram of the correlation between *CDH1* expression and *VIM* expression. The red line was a fitting curve. Pearson correlation coefficient was used for correlation analysis. (M) Schematic diagram of the correlation between *CDH1* expression and *TWIST2* expression. The red line was a fitting curve. Pearson correlation coefficient was used for correlation analysis.

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Additional information

Author Contributions

H.J. and F.L. conceived the idea and designed the analysis. X.C. performed all bioinformatics analyses. Y.Y., X.L., and L.C. provided technical assistance and helpful comments. H.J., F.L. and X.C. wrote the manuscript. All authors approved the final version.

Online content

Code availability are available at *https://github.com/XinleiCai/MEMORY*.

Additional files

Supplementary tables

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Reviewer #1 (Public review):

Summary:

The authors propose a new technique which they name "Multi-gradient Permutation Survival Analysis (MEMORY)" that they use to identify "Genes Steadily Associated with Prognosis (GEARs)" using RNA-seq data from the TCGA database. The contribution of this method is one of the key stated aims of the paper. The majority of the paper focuses on various downstream analyses that make use of the specific GEARs identified by MEMORY to derive biological insights, with a particular focus on lung adenocarcinoma (LUAD) and breast invasive carcinoma (BRCA) which are stated to be representative of other cancers and are observed to have enriched mitosis and immune signatures, respectively. Through the lens of these cancers, these signatures are the focus of significant investigation in the paper.

Strengths:

The approach for MEMORY is well-defined and clearly presented, albeit briefly. This affords statisticians and bioinformaticians the ability to effectively scrutinize the proposed methodology and may lead to further advancements in this field. The scientific aspects of the paper (e.g., the results based on the use of MEMORY and the downstream bioinformatics workflows) are conveyed effectively and in a way that is digestible to an individual that is not deeply steeped in the cancer biology field.

Weaknesses:

Comparatively little of the paper is devoted to the justification of MEMORY (i.e., the authors' method) for identification of genes that are important broadly for the understanding of cancer. The authors' approach is explained in the methods section of the paper, but no comparison or reference is made to any other methods that have been developed for similar purposes, and no results are shown to illustrate the robustness of the proposed method (e.g., is it sensitive to subtle changes in how it is implemented).

For example, in the first part of the MEMORY algorithm, gene expression values are dichotomized at the sample median, and a log-rank test is performed. This would seemingly result in an unnecessary loss of information for detecting an association between gene expression and survival. Moreover, while dichotomizing gene expressions at the median is optimal from an information theory perspective (i.e., it creates equally sized groups), there is no reason to believe that median-dichotomization is correct vis-à-vis the relationship between gene expression and survival. If a gene really matters and expression only differentiates survival more towards the tail of the empirical gene expression distribution, median-dichotomization could dramatically lower power to detect group-wise differences. Notwithstanding this point, the reviewer acknowledges that dichotomization offers a straightforward approach to model gene expression and is widely used. This approach is nonetheless an example of a limitation of the current version of MEMORY that could be addressed to improve the methodology.

If I understand correctly, for each cancer the authors propose to search for the smallest subsample size (i.e., the smallest value of k_{j}) were there is at least one gene with a survival analysis p-value <0.05 for each of the 1000 sampled datasets. Then, any gene with a p-value <0.05 in 80% of the 1000 sampled datasets would be called a GEAR for that cancer. The 80% value here is arbitrary but that is a minor point. I acknowledge that something must be chosen.

Presumably the gene with the largest effect for the cancer will define the value of K_{j} and, if the effect is large, this may result in other genes with smaller effects not being defined as a GEAR for that cancer by virtue of the 80% threshold. Thus, a gene being a GEAR is related to the strength of association for other genes in addition to its own strength of association. One could imagine that a gene that has a small-to-moderate effect consistently across many cancers may not show up as GEAR in any of them (if there are [potentially different] genes with more substantive effects for those cancers). Is this desirable?

The term "steadily associated" implies that a signal holds up across subsample gradients. Effectively this makes the subsampling a type of indirect adjustment to ensure the evidence of association is strong enough. How well this procedure performs in repeated use (i.e., as a statistical procedure) is not clear.

Assuredly subsampling sets the bar higher than requiring a nominal p-value to be beneath the 0.05 threshold based on analysis of the full data set. The author's note that the MEMORY has several methodological limitations, "chief among them is the need for rigorous, large-scale multiple-testing adjustment before any GEAR list can be considered clinically actionable." The reviewer agrees and would add that it may be difficult to address this limitation within the author's current framework. Moreover, should the author's method be used before such corrections are available given their statement? Perhaps clarification of what it means to be clinically actionable could help here. If a researcher uses MEMORY to screen for GEARs based on the current methodology, what do the authors recommend be done to select a subset of GEARs worthy of additional research/investment?

https://doi.org/10.7554/eLife.101619.2.sa3

Reviewer #2 (Public review):

Summary:

The authors are trying to come up with a list of genes (GEAR genes) that are consistently associated with cancer patient survival based on TCGA database. A method named "Multi-gradient Permutation Survival Analysis" was created based on bootstrapping and gradually increasing the sample size of the analysis. Only the genes with consistent performance in this analysis process are chosen as potential candidates for further analyses.

Strengths:

The authors describe in details their proposed method and the list of the chosen genes from the analysis. Scientific meaning and potential values of their findings are discussed in the context of published results in this field.

Weaknesses:

Some steps of the proposed method (especially the definition survival analysis similarity (SAS) need further clarification or details since it would be difficult if anyone tries to reproduce the results.

If the authors can improve the clarity of the manuscript, including the proposed method and there is no major mistake there, the proposed approach can be applied to other diseases (assuming TCGA type of data is available for them) to identify potential gene lists, based on which drug screening can be performed to identify potential target for development.

https://doi.org/10.7554/eLife.101619.2.sa2

Reviewer #4 (Public review):

Thank you to the authors for their detailed responses and changes in relation to my questions. They have addressed all my concerns around methodological and inference clarity. I would still recommend against the use of feature/pathway selection techniques where there is no way of applying formal error control. I am pleased to read, however, that the authors are planning to develop this in future work. My edited review reflects these changes:

The authors apply what I gather is a novel methodology titled "Multi-gradient Permutation Survival Analysis" to identify genes that are robustly associated with prognosis ("GEARs") using tumour expression data from 15 cancer types available in the TCGA. The resulting lists of GEARs are then interrogated for biological insights using a range of techniques including connectivity and gene enrichment analysis.

I reviewed this paper primarily from a statistical perspective. Evidently an impressive amount of work has been conducted, concisely summarised, and great effort has been undertaken to add layers of insight to the findings. I am no stranger to what an undertaking this would have been. My primary concern, however, is that the novel statistical procedure proposed, and applied to identify the gene lists, as far as I can tell offers no statistical error control nor quantification. Consequently we have no sense what proportion of the highlighted GEAR genes and networks are likely to just be noise.

Major comments:

The main methodology used to identify the GEAR genes, "Multi-gradient Permutation Survival Analysis" does not formally account for multiple testing and offers no formal error control. Meaning we are left without knowing what the family wise (aka type 1) error rate is among the GEAR lists, nor the false discovery rate. I appreciate the emphasis on reproducibility, but I would generally recommend against the use of any feature selection methodology which does not provide error quantification because otherwise we do not know if we are encouraging our colleagues and/or readers to put resource into lists of genes that contain more noise than not. I am glad though and appreciative that the authors intend to develop this in future work.

The authors make a good point that, despite lack of validation in an external independent dataset, it is still compelling work given the functional characterisation and literature validation. I am pleased though that the authors agree validation in an independent dataset is an important next step, and plan to do so in future work.

https://doi.org/10.7554/eLife.101619.2.sa1

Author response:

The following is the authors’ response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The authors propose a new technique which they name "Multi-gradient Permutation Survival Analysis (MEMORY)" that they use to identify "Genes Steadily Associated with Prognosis (GEARs)" using RNA-seq data from the TCGA database. The contribution of this method is one of the key stated aims of the paper. The vast majority of the paper focuses on various downstream analyses that make use of the specific GEARs identified by MEMORY to derive biological insights, with a particular focus on lung adenocarcinoma (LUAD) and breast invasive carcinoma (BRCA) which are stated to be representative of other cancers and are observed to have enriched mitosis and immune signatures, respectively. Through the lens of these cancers, these signatures are the focus of significant investigation in the paper.*
>
> *Strengths:*
>
> *The approach for MEMORY is well-defined and clearly presented, albeit briefly. This affords statisticians and bioinformaticians the ability to effectively scrutinize the proposed methodology and may lead to further advancements in this field.*
>
> *The scientific aspects of the paper (e.g., the results based on the use of MEMORY and the downstream bioinformatics workflows) are conveyed effectively and in a way that is digestible to an individual who is not deeply steeped in the cancer biology field.*
>
> *Weaknesses:*
>
> *I was surprised that comparatively little of the paper is devoted to the justification of MEMORY (i.e., the authors' method) for the identification of genes that are important broadly for the understanding of cancer. The authors' approach is explained in the methods section of the paper, but no rationale is given for why certain aspects of the method are defined as they are. Moreover, no comparison or reference is made to any other methods that have been developed for similar purposes and no results are shown to illustrate the robustness of the proposed method (e.g., is it sensitive to subtle changes in how it is implemented).*
>
> *For example, in the first part of the MEMORY algorithm, gene expression values are dichotomized at the sample median and a log-rank test is performed. This would seemingly result in an unnecessary loss of information for detecting an association between gene expression and survival. Moreover, while dichotomizing at the median is optimal from an information theory perspective (i.e., it creates equally sized groups), there is no reason to believe that median-dichotomization is correct vis-à-vis the relationship between gene expression and survival. If a gene really matters and expression only differentiates survival more towards the tail of the empirical gene expression distribution, median-dichotomization could dramatically lower the power to detect group-wise differences.*

Thanks for these valuable comments!! We understand the reviewer’s concern regarding the potential loss of information caused by median-based dichotomization. In this study, we adopted the median as the cut-off value to stratify gene expression levels primarily for the purpose of data balancing and computational simplicity. This approach ensures approximately equal group sizes, which is particularly beneficial in the context of limited sample sizes and repeated sampling. While we acknowledge that this method may discard certain expression nuances, it remains a widely used strategy in survival analysis. To further evaluate and potentially enhance sensitivity, alternative strategies such as percentile-based cutoffs or survival models using continuous expression values (e.g., Cox regression) may be explored in future optimization of the MEMORY pipeline. Nevertheless, we believe that this dichotomization approach offers a straightforward and effective solution for the initial screening of survival-associated genes. We have now included this explanation in the revised manuscript (Lines 391–393).

> *Specifically, the authors' rationale for translating the Significant Probability Matrix into a set of GEARs warrants some discussion in the paper. If I understand correctly, for each cancer the authors propose to search for the smallest sample size (i.e., the smallest value of k_{j}) were there is at least one gene with a survival analysis p-value <0.05 for each of the 1000 sampled datasets. I base my understanding on the statement "We defined the sampling size k_{j} reached saturation when the max value of column j was equal to 1 in a significant-probability matrix. The least value of k_{j} was selected". Then, any gene with a p-value <0.05 in 80% of the 1000 sampled datasets would be called a GEAR for that cancer. The 80% value here seems arbitrary but that is a minor point. I acknowledge that something must be chosen. More importantly, do the authors believe this logic will work effectively in general? Presumably, the gene with the largest effect for a cancer will define the value of K_{j}, and, if the effect is large, this may result in other genes with smaller effects not being selected for that cancer by virtue of the 80% threshold. One could imagine that a gene that has a small-tomoderate effect consistently across many cancers may not show up as a gear broadly if there are genes with more substantive effects for most of the cancers investigated. I am taking the term "Steadily Associated" very literally here as I've constructed a hypothetical where the association is consistent across cancers but not extremely strong. If by "Steadily Associated" the authors really mean "Relatively Large Association", my argument would fall apart but then the definition of a GEAR would perhaps be suboptimal. In this latter case, the proposed approach seems like an indirect way to ensure there is a reasonable effect size for a gene's expression on survival.*

Thank you for the comment and we apologize for the confusion! A_{ij} refers to the value of gene i under gradient j in the significant-probability matrix, primarily used to quantify the statistical probability of association with patient survival for ranking purposes. We believe that GEARs are among the top-ranked genes, but there is no established metric to define the optimal threshold. An 80% threshold is previously employed as an empirical standard in studies related to survival estimates [1]. In addition, we acknowledge that the determination of the saturation point k_j is influenced by the earliest point at which any gene achieves consistent significance across 1000 permutations. We recognize that this may lead to the under representation of genes with moderate but consistent effects, especially in the presence of highly significant genes that dominate the statistical landscape. We therefore empirically used $A_{ij} > 0.8$ the threshold to distinguish between GEARs and non-GEARs. Of course, this parameter variation may indeed result in the loss of some GEARs or the inclusion of non-GEARs. We also agree that future studies could investigate alternative metrics and more refined thresholds to improve the application of GEARs.

Regarding the term ‘Steadily Associated’, we define GEARs based on statistical robustness across subsampled survival analyses within individual cancer types, rather than cross-cancer consistency or pan-cancer moderate effects. Therefore, our operational definition of “steadiness” emphasizes within-cancer reproducibility across sampling gradients, which does not necessarily exclude high-effect-size genes. Nonetheless, we agree that future extensions of MEMORY could incorporate cross-cancer consistency metrics to capture genes with smaller but reproducible pan-cancer effects.

> *The paper contains numerous post-hoc hypothesis tests, statements regarding detected associations and correlations, and statements regarding statistically significant findings based on analyses that would naturally only be conducted in light of positive results from analyses upstream in the overall workflow. Due to the number of statistical tests performed and the fact that the tests are sometimes performed using data-driven subgroups (e.g., the mitosis subgroups), it is highly likely that some of the findings in the work will not be replicable. Of course, this is exploratory science, and is to be expected that some findings won't replicate (the authors even call for further research into key findings). Nonetheless, I would encourage the authors to focus on the quantification of evidence regarding associations or claims (i.e., presenting effect estimates and uncertainty intervals), but to avoid the use of the term statistical significance owing to there being no clear plan to control type I error rates in any systematic way across the diverse analyses there were performed.*

Thank you for the comment! We agree that rigorous control of type-I error is essential once a definitive list of prognostic genes is declared. The current implementation of MEMORY, however, is deliberately positioned as an exploratory screening tool: each gene is evaluated across 10 sampling gradients and 1,000 resamples per gradient, and the only quantity carried forward is its reproducibility probability (A_{ij}).

Because these probabilities are derived from aggregate "votes" rather than single-pass P-values, the influence of any one unadjusted test is inherently diluted. In another words, whether or not a per-iteration BH adjustment is applied does not materially affect the ranking of genes by reproducibility, which is the key output at this stage. However, we also recognize that a clinically actionable GEARs catalogue will require extensive, large-scale multiple-testing adjustments. Accordingly, future versions of MEMORY will embed a dedicated false-positive control framework tailored to the final GEARs list before any translational application. We have added this point in the 'Discussion' in the revised manuscript (Lines 350-359).

> *A prespecified analysis plan with hypotheses to be tested (to the extent this was already produced) and a document that defines the complete scope of the scientific endeavor (beyond that which is included in the paper) would strengthen the contribution by providing further context on the totality of the substantial work that has been done. For example, the focus on LUAD and BRCA due to their representativeness could be supplemented by additional information on other cancers that may have been investigated similarly but where results were not presented due to lack of space.*

We thank the reviewer for requesting greater clarity on the analytic workflow. The MEMORY pipeline was fully specified before any results were examined and is described in 'Methods' (Lines 386–407). By contrast, the pathway-enrichment and downstream network/mutation analyses were deliberately exploratory: their exact content necessarily depended on which functional categories emerged from the unbiased GEAR screen.

Our screen revealed a pronounced enrichment of mitotic signatures in LUAD and immune signatures in BRCA.

We then chose these two cancer types for deeper "case-study" analysis because they contained the largest sample sizes among all cancers showing mitotic- or immune-dominated GEAR profiles, and provided the greatest statistical power for follow-up investigations. We have added this explanation into the revised manuscript (Line 163, 219-220).

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *The authors are trying to come up with a list of genes (GEAR genes) that are consistently associated with cancer patient survival based on TCGA database. A method named "Multi-gradient Permutation Survival Analysis" was created based on bootstrapping and gradually increasing the sample size of the analysis. Only the genes with consistent performance in this analysis process are chosen as potential candidates for further analyses.*
>
> *Strengths:*
>
> *The authors describe in detail their proposed method and the list of the chosen genes from the analysis. The scientific meaning and potential values of their findings are discussed in the context of published results in this field.*
>
> *Weaknesses:*
>
> *Some steps of the proposed method (especially the definition of survival analysis similarity (SAS) need further clarification or details since it would be difficult if anyone tries to reproduce the results. In addition, the multiplicity (a large number of p-values are generated) needs to be discussed and/or the potential inflation of false findings needs to be part of the manuscript.*

Thank you for the reviewer's insightful comments. Accordingly, in the revised manuscript, we have provided a more detailed explanation of the definition and calculation of Survival-Analysis Similarity (SAS) to ensure methodological clarity and reproducibility (Lines 411-428); and the full code is now publicly available on GitHub (https://github.com/XinleiCai/MEMORY). We have also expanded the 'Discussion' to clarify our position on false-positive control: future releases of MEMORY will incorporate a dedicated framework to control false discoveries in the final GEARs catalogue, where itself will be subjected to rigorous, large-scale multiple-testing adjustment.

> *If the authors can improve the clarity of the proposed method and there is no major mistake there, the proposed approach can be applied to other diseases (assuming TCGA type of data is available for them) to identify potential gene lists, based on which drug screening can be performed to identify potential target for development.*

Thank you for the suggestion. All source code has now been made publicly available on GitHub for reference and reuse. We agree that the GEAR lists produced by MEMORY hold considerable promise for drugscreening and target-validation efforts, and the framework could be applied to any disease with TCGA-type data. Of course, we also notice that the current GEAR catalogue should first undergo rigorous, large-scale multipletesting correction to further improve its precision before broader deployment.

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *The authors describe a valuable method to find gene sets that may correlate with a patient's survival. This method employs iterative tests of significance across randomised samples with a range of proportions of the original dataset. Those genes that show significance across a range of samples are chosen. Based on these gene sets, hub genes are determined from similarity scores.*

> *Strengths:*
>
> *MEMORY allows them to assess the correlation between a gene and patient prognosis using any available transcriptomic dataset. They present several follow-on analyses and compare the gene sets found to previous studies.*
>
> *Weaknesses:*
>
> *Unfortunately, the authors have not included sufficient details for others to reproduce this work or use the MEMORY algorithm to find future gene sets, nor to take the gene findings presented forward to be validated or used for future hypotheses.*

Thank you for the reviewer's comments! We apologize for the inconvenience and the lack of details.

Followed the reviewer's valuable suggestion, we have now made all source code and relevant scripts publicly available on GitHub to ensure full reproducibility and facilitate future use of the MEMORY algorithm for gene discovery and hypothesis generation.

> ***Reviewer #4 (Public review):***
>
> *The authors apply what I gather is a novel methodology titled "Multi-gradient Permutation Survival Analysis" to identify genes that are robustly associated with prognosis ("GEARs") using tumour expression data from 15 cancer types available in the TCGA. The resulting lists of GEARs are then interrogated for biological insights using a range of techniques including connectivity and gene enrichment analysis.*
>
> *I reviewed this paper primarily from a statistical perspective. Evidently, an impressive amount of work has been conducted, and concisely summarised, and great effort has been undertaken to add layers of insight to the findings. I am no stranger to what an undertaking this would have been. My primary concern, however, is that the novel statistical procedure proposed, and applied to identify the gene lists, as far as I can tell offers no statistical error control or quantification. Consequently, we have no sense of what proportion of the highlighted GEAR genes and networks are likely to just be noise.*
>
> *Major comments:*
>
> *(1) The main methodology used to identify the GEAR genes, "Multi-gradient Permutation Survival Analysis" does not formally account for multiple testing and offers no formal error control. Meaning we are left with no understanding of what the family-wise (aka type 1) error rate is among the GEAR lists, nor the false discovery rate. I would generally recommend against the use of any feature selection methodology that does not provide some form of error quantification and/or control because otherwise we do not know if we are encouraging our colleagues and/or readers to put resources into lists of genes that contain more noise than not. There are numerous statistical techniques available these days that offer error control, including for lists of p-values from arbitrary sets of tests (see expansion on this and some review references below).*

Thank you for your thoughtful and important comment! We fully agree that controlling type I error is critical when identifying gene sets for downstream interpretation or validation. As an exploratory study, our primary aim was to define and screen for GEARs by using the MEMORY framework; however, we acknowledge that the current implementation of MEMORY does not include a formal procedure for error control. Given that MEMORY relies on repeated sampling and counts the frequency of statistically significant p-values, applying standard p-value–based multiple-testing corrections at the individual test level would not meaningfully reduce the false-positive rate in this framework.

We believe that error control should instead be applied at the level of the final GEAR catalogue. However, we also recognize that conventional correction methods are not directly applicable. In future versions of MEMORY, we plan to incorporate a dedicated and statistically appropriate false-positive control module tailored specifically to the aggregated outputs of the pipeline. We have clarified this point explicitly in the revised manuscript. (Lines 350-359)

> *(2) Similarly, no formal significance measure was used to determine which of the strongest "SAS" connections to include as edges in the "Core Survival Network".*

We agree that the edges in the Core Survival Network (CSN) were selected based on the top-ranked SAS values rather than formal statistical thresholds. This was a deliberate design choice, as the CSN was intended as a heuristic similarity network to prioritize genes for downstream molecular classification and biological exploration, not for formal inference. To address potential concerns, we have clarified this intent in the revised manuscript, and we now explicitly state that the network construction was based on empirical ranking rather than statistical significance (Lines 422-425).

> *(3) There is, as far as I could tell, no validation of any identified gene lists using an independent dataset external to the presently analysed TCGA data.*

Thank you for the comment. We acknowledge that no independent external dataset was used in the present study to validate the GEARs lists. However, the primary aim of this work was to systematically identify and characterize genes with robust prognostic associations across cancer types using the MEMORY framework. To assess the biological relevance of the resulting GEARs, we conducted extensive downstream analyses including functional enrichment, mutation profiling, immune infiltration comparison, and drug-response correlation. These analyses were performed across multiple cancer types and further supported by a wide range of published literature.

We believe that this combination of functional characterization and literature validation provides strong initial support for the robustness and relevance of the GEARs lists. Nonetheless, we agree that validation in independent datasets is an important next step, and we plan to carry this out in future work to further strengthen the clinical application of MEMORY.

> *(4) There are quite a few places in the methods section where descriptions were not clear (e.g. elements of matrices referred to without defining what the columns and rows are), and I think it would be quite challenging to re-produce some aspects of the procedures as currently described (more detailed notes below).*

We apologize for the confusion. In the revised manuscript, we have provided a clearer and more detailed description of the computational workflow of MEMORY to improve clarity and reproducibility.

> *(5) There is a general lack of statistical inference offered. For example, throughout the gene enrichment section of the results, I never saw it stated whether the pathways highlighted are enriched to a significant degree or not.*

We apologize for not clearly stating this information in the original manuscript. In the revised manuscript, we have updated the figure legend to explicitly report the statistical significance of the enriched pathways (Line 870, 877, 879-880).

> ***Reviewer #1 (Recommendations for the authors):***
>
> *Overall, the paper reads well but there are numerous small grammatical errors that at times cost me non-trivial amounts of time to understand the authors' key messages.*

We apologize for the grammatical errors that hindered clarity. In response, we have thoroughly revised the manuscript for grammar, spelling, and overall language quality.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *Major comments:*
>
> *(1) Line 427: survival analysis similarity (SAS) definition. Any reference on this definition and why it is defined this way? Can the SAS value be negative? Based on line 429 definition, if A and B are exactly the same, SAS ~ 1; completely opposite, SAS =0; otherwise, SAS could be any value, positive or negative. So it is hard to tell what SAS is measuring. It is important to make sure SAS can measure the similarity in a systematic and consistent way since it is used as input in the following network analysis.*

We apologize for the confusion caused by the ambiguity in the original SAS formula. The SAS metric was inspired by the Jaccard index, but we modified the denominator to increase contrast between gene pairs. Specifically, the numerator counts the number of permutations in which both genes are simultaneously significant (i.e., both equal to 1), while the denominator is the sum of the total number of significant events for each gene minus twice the shared significant count. An additional +1 term was included in the denominator to avoid division by zero. This formulation ensures that SAS is always non-negative and bounded between 0 and 1, with higher values indicating greater similarity. We have clarified this definition and updated the formula in the revised manuscript (Lines 405-425).

> *(2) For the method with high dimensional data, multiplicity adjustment needs to be discussed, but it is missing in the manuscript. A 5% p-value cutoff was used across the paper, which seems to be too liberal in this type of analysis. The suggestion is to either use a lower cutoff value or use False Discovery Rate (FDR) control methods for such adjustment. This will reduce the length of the gene list and may help with a more focused discussion.*

We appreciate the reviewer's suggestion regarding multiplicity. MEMORY is intentionally positioned as an exploratory screen: each gene is tested across 10 sampling gradients and 1,000 resamples, and only its reproducibility probability (A_{ij}) is retained. Because this metric is an aggregate of 1,000 "votes" the influence of any single unadjusted P-value is already strongly diluted; adding a per-iteration BH/FDR step therefore has negligible impact on the reproducibility ranking that drives all downstream analyses.

That said, we recognize that a clinically actionable GEARs catalogue must undergo formal, large-scale multipletesting correction. Future releases of MEMORY will incorporate an error control module applied to the consolidated GEAR list before any translational use. We have now added a statement to this effect in the revised manuscript (Lines 350-359).

> *(3) To allow reproducibility from others, please include as many details as possible (software, parameters, modules etc.) for the analyses performed in different steps.*

All source codes are now publically available on GitHub. We have also added the GitHub address in the section Online Content.

> *Minor comments or queries:*

> *(4) The manuscript needs to be polished to fix grammar, incomplete sentences, and missing figures.*

Thank you for the suggestion. We have thoroughly proofread the manuscript to correct grammar, complete any unfinished sentences, and restore or renumber all missing figure panels. All figures are now properly referenced in the text.

> *(5) Line 131: "survival probability of certain genes" seems to be miss-leading. Are you talking about its probability of associating with survival (or prognosis)?*

Sorry for the oversight. What we mean is the probability that a gene is found to be significantly associated with survival across the 1,000 resamples. We have revised the statement to “significant probability of certain genes” (Line 102).

> *(6) Lines 132, 133: "remained consistent": the score just needs to stay > 0.8 as the sample increases, or the score needs to be monotonously non-decreasing?*

We mean the score stay above 0.8. We understand “remained consistent” is confusing and now revised it to “remained above 0.8”.

> *(7) Lines 168-170 how can supplementary figure 5A-K show "a certain degree of correlation with cancer stages"?*

Sorry for the confusion! We have now revised Supplementary Figure 5A–K to support the visual impression with formal statistics. For each cancer type, we built a contingency table of AJCC stage (I–IV) versus hub-gene subgroup (Low, Mid, High) and applied Pearson’s x^2 test (Monte-Carlo approximation, 10^5 replicates when any expected cell count < 5). The x^2 statistic and p-value are printed beneath every panel; eight of the eleven cancers show a significant association (p-value < 0.05), while LUSC, THCA and PAAD do not.We have replaced the vague phrase “a certain degree of correlation” with this explicit statistical statement in the revised manuscript (Lines 141-143).

> *(8) Lines 172-174: since the hub genes are a subset of GEAR genes through CSN construction, it is not a surprise of the consistency. any explanation about PAAD that is shown only in GOEA with GEARs but not with hub genes?*

Thanks for raising this interesting point! In PAAD the Core Survival Network is unusually diffuse: the top-ranked SAS edges are distributed broadly rather than converging on a single dense module. Because of this flat topology, the ten highest-degree nodes (our hub set) do not form a tightly interconnected cluster, nor are they collectively enriched in the mitosis-related pathway that dominates the full GEAR list. This might explain that the mitotic enrichment is evident when all PAAD GEARs were analyzed but not when the analysis is confined to the far smaller—and more functionally dispersed—hub-gene subset.

> *(9) Lines 191: how the classification was performed? Tool? Cutoff values etc?*

The hub-gene-based molecular classification was performed in R using hierarchical clustering. Briefly, we extracted the $log_2(TPM+1)$ expression matrix of hub genes, computed Euclidean distances between samples, and applied Ward’s minimum variance method (hclust, method = "ward.D2"). The resulting dendrogram was then divided into three groups (cutree, k = 3), corresponding to low, mid, and high expression classes. These parameters were selected based on visual inspection of clustering structure across cancer types. We have added this information to the revised ‘Methods’ section (Lines 439-443).

> *(10) Lines 210-212: any statistics to support the conclusion? The bar chat of Figure 3B seems to support that all mutations favor ML & MM.*

We agree that formal statistical support is important for interpreting groupwise comparisons. In this case, however, several of the driver events, such as ROS1 and ERBB2, had very small subgroup counts, which violate the assumptions of Pearson's x^2 test. While we explored x^2 and Fisher's exact tests, the results were unstable due to sparse counts. Therefore, we chose to present these distributions descriptively to illustrate the observed subtype preferences across different driver mutations (Figure 3B). We have revised the manuscript text to clarify this point (Lines 182-188).

> *(11) Line 216: should supplementary Figure 6H-J be "6H-I"?*

We apologize for the mistake. We have corrected it in the revised manuscript.

> *(12) Line 224: incomplete sentence starting with "To further the functional... ".*

Thanks! We have made the revision and it states now "To further expore the functional implications of these mutations, we enriched them using a pathway system called Nested Systems in Tumors (NeST)".

> *(13) Lines 261-263: it is better to report the median instead of the mean. Use log scale data for analysis or use non-parametric methods due to the long tail of the data.*

Thank you for the very helpful suggestion. In the revised manuscript, we now report the median instead of the mean to better reflect the distribution of the data. In addition, we have applied log-scale transformation where appropriate and replaced the original statistical tests with non-parametric Wilcoxon ranksum tests to account for the long-tailed distribution. These changes have been implemented in both the main text and figure legends (Lines 234–237, Figure 5F).

> *(14) Line 430: why based on the first sampling gradient, i.e. k_1 instead of the k_j selected? Or do you mean k_j here?*

Thanks for this question! We deliberately based SAS on the vectors from the first sampling gradient (k_1, ≈ 10 % of the cohort). At this smallest sample size, the binary significance patterns still contain substantial variation, and many genes are not significant in every permutation. Based on this, we think the measure can meaningfully identify gene pairs that behave concordantly throughout the gradient permutation.

We have now added a sentence to clarify this in the Methods section (Lines 398–403).

> *(15) Need clarification on how the significant survival network was built.*

Thank you for pointing this out. We have now provided a more detailed clarification of how the Survival-Analysis Similarity (SAS) metric was defined and applied in constructing the core survival network (CSN), including the rationale for key parameter choices (Lines 409–430). Additionally, we have made full source code publicly available on GitHub to facilitate transparency and reproducibility (https://github.com/XinleiCai/MEMORY).

> *(16) Line 433: what defines the "significant genes" here? Are they the same as GEAR genes? And what are total genes, all the genes?*

We apologize for the inconsistency in terminology, which may have caused confusion. In this context,

“significant genes” refers specifically to the GEARs (Genes Steadily Associated with Prognosis). The SAS values were calculated between each GEAR and all genes. We have revised the manuscript to clarify this by consistently using the term “GEARs” throughout.

> *(17) Line 433: more detail on how SAS values were used will be helpful. For example, were pairwise SAS values fed into Cytoscape as an additional data attribute (on top of what is available in TCGA) or as the only data attribute for network building?*

The SAS values were used as the sole metric for defining connections (edges) between genes in the construction of the core survival network (CSN). Specifically, we calculated pairwise SAS values between each GEAR and all other genes, then selected the top 1,000 gene pairs with the highest SAS scores to construct the network. No additional data attributes from TCGA (such as expression levels or clinical features) were used in this step. These selected pairs were imported into Cytoscape solely based on their SAS values to visualize the CSN.

> *(18) Line 434: what is "ranking" here, by degree? Is it the same as "nodes with top 10 degrees" at line 436?*

The “ranking” refers specifically to the SAS values between gene pairs. The top 1,000 ranked SAS values were selected to define the edges used in constructing the Core Survival Network (CSN).

Once the CSN was built, we calculated the degree (number of connections) for each node (i.e., each gene). The

“top 10 degrees” mentioned on Line 421 refers to the 10 genes with the highest node degrees in the CSN. These were designated as hub genes for downstream analyses.

We have clarified this distinction in the revised manuscript (Line 398-403).

> *(19) Line 435: was the network built in Cytoscape? Or built with other tool first and then visualized in Cytoscape?*

The network was constructed in R by selecting the top 1,000 gene pairs with the highest SAS values to define the edges. This edge list was then imported into Cytoscape solely for visualization purposes. No network construction or filtering was performed within Cytoscape itself. We have clarified this in the revised ‘Methods’ section (Lines 424-425).

> *(20) Line 436: the degree of each note was calculated, what does it mean by "degree" here and is it the same as the number of edges? How does it link to the "higher ranked edges" in Line 165?*

The “degree” of a node refers to the number of edges connected to that node—a standard metric in graph theory used to quantify a node’s centrality or connectivity in the network. It is equivalent to the number of edges a gene shares with others in the CSN.

The “higher-ranked edges” refer to the top 1,000 gene pairs with the highest SAS values, which we used to construct the Core Survival Network (CSN). The degree for each node was computed within this fixed network, and the top 10 nodes with the highest degree were selected as hub genes. Therefore, the node degree is largely determined by this pre-defined edge set.

> *(21) Line 439: does it mean only 1000 SAS values were used or SAS values from 1000 genes, which should come up with 1000 choose 2 pairs (~ half million SAS values).*

We computed the SAS values between each GEAR gene and all other genes, resulting in a large number of pairwise similarity scores. Among these, we selected the top 1,000 gene pairs with the highest SAS values—regardless of how many unique genes were involved—to define the edges in the Core Survival Network (CSN). In another words, the network is constructed from the top 1,000 SAS-ranked gene pairs, not from all possible combinations among 1,000 genes (which would result in nearly half a million pairs). This approach yields a sparse network focused on the strongest co-prognostic relationships.

We have clarified this in the revised 'Methods' section (Lines 409–430).

> *(22) Line 496: what tool is used and what are the parameters set for hierarchical clustering if someone would like to reproduce the result?*

The hierarchical clustering was performed in R using the hclust function with Ward's minimum variance method (method = "ward.D2"), based on Euclidean distance computed from the log-transformed expression matrix ($log_2(TPM+1)$). Cluster assignment was done using the cutree function with k = 3 to define low, mid, and high expression subgroups. These settings have now been explicitly stated in the revised 'Methods' section (Lines 439–443) to facilitate reproducibility.

> *(23) Lines 901-909: Figure 4 missing panel C. Current panel C seems to be the panel D in the description.*

Sorry for the oversights and we have now made the correction (Line 893).

> *(24) Lines 920-928: Figure 6C: considering a higher bar to define "significant".*

We agree that applying a more stringent cutoff (e.g., $p < 0.01$) may reduce potential false positives. However, given the exploratory nature of this study, we believe the current threshold remains appropriate for the purpose of hypothesis generation.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) The title says the genes that are "steadily" associated are identified, but what you mean by the word "steadily" is not defined in the manuscript. Perhaps this could mean that they are consistently associated in different analyses, but multiple analyses are not compared.*

In our manuscript, "steadily associated" refers to genes that consistently show significant associations with patient prognosis across multiple sample sizes and repeated resampling within the MEMORY framework (Lines 65–66). Specifically, each gene is evaluated across 10 sampling gradients (from ~10% to 100% of the cohort) with 1,000 permutations at each level. A gene is defined as a GEAR if its probability of being significantly associated with survival remains ≥ 0.8 throughout the whole permutation process. This stability in signal under extensive resampling is what we refer to as "steadily associated."

> *(2) I think the word "gradient" is not appropriately used as it usually indicates a slope or a rate of change. It seems to indicate a step in the algorithm associated with a sampling proportion.*

Thank you for pointing out the potential ambiguity in our use of the term “gradient.” In our study, we used “gradient” to refer to stepwise increases in the sample proportion used for resampling and analysis. We have now revised it to “progressive”.

> *(3) Make it clear that the name "GEARs" is introduced in this publication.*

Done.

> *(4) Sometimes the document is hard to understand, for example, the sentence, "As the number of samples increases, the survival probability of certain genes gradually approaches 1." It does not appear to be calculating "gene survival probability" but rather a gene's association with patient survival. Or is it that as the algorithm progresses genes are discarded and therefore do have a survival probability? It is not clear.*

What we intended to describe is the probability that a gene is judged significant in the 1,000 resamples at a given sample-size step, that is, its reproducibility probability in the MEMORY framework. We have now revised the description (Lines 101-104).

> *(5) The article lacks significant details, like the type of test used to generate p-values. I assume it is the log-rank test from the R survival package. This should be explicitly stated. It is not clear why the survminer R package is required or what function it has. Are the p-values corrected for multiple hypothesis testing at each sampling?*

We apologize for the lack of details. In each sampling iteration, we used the log-rank test (implemented via the survdiff function in the R survival package) to evaluate the prognostic association of individual genes. This information has now been explicitly added to the revised manuscript.

The survminer package was originally included for visualization purposes, such as plotting illustrative Kaplan– Meier curves. However, since it did not contribute to the core statistical analysis, we have now removed this package from the Methods section to avoid confusion (Lines 386-407).

As for multiple-testing correction, we did not adjust p-values in each iteration, because the final selection of GEARs is based on the frequency with which a gene is found significant across 1,000 resamples (i.e., its reproducibility probability). Classical FDR corrections at the per-sample level do not meaningfully affect this aggregate metric. That said, we fully acknowledge the importance of multiple-testing control for the final GEARs catalogue. Future versions of the MEMORY framework will incorporate appropriate adjustment procedures at that stage.

> *(6) It is not clear what the survival metric is. Is it overall survival (OS) or progression-free survival (PFS), which would be common choices?*

It’s overall survival (OS).

> *(7) The treatment of the patients is never considered, nor whether the sequencing was performed pre or posttreatment. The patient's survival will be impacted by the treatment that they receive, and many other factors like commodities, not just the genomics.*

We initially thought there exist no genes significantly associated with patient survival (GEARs) without counting so many different influential factors. This is exactly what motivated us to invent the

MEMORY. However, this work proves “we were wrong”, and it demonstrates the real power of GEARs in determining patient survival. Of course, we totally agree with the reviewer that incorporating therapy variables and other clinical covariates will further improve the power of MEMORY analyses.

> *(8) As a paper that introduces a new analysis method, it should contain some comparison with existing state of the art, or perhaps randomised data.*

Our understanding is --- the MEMORY presents as an exploratory and proof-of-concept framework. Comparison with regular survival analyses seems not reasonable. We have added some discussion in revised manuscript (Lines 350-359).

> *(9) In the discussion it reads, "it remains uncertain whether there exists a set of genes steadily associated with cancer prognosis, regardless of sample size and other factors." Of course, there are many other factors that may alter the consistency of important cancer genes, but sample size is not one of them. Sample size merely determines whether your study has sufficient power to detect certain gene effects, it does not effect whether genes are steadily associated with cancer prognosis in different analyses. (Of course, this does depend on what you mean by "steadily".)*

We totally agree with reviewer that sample size itself does not alter a gene’s biological association with prognosis; it only affects the statistical power to detect that association. Because this study is exploratory and we were initially uncertain whether GEARs existed, we first examined the impact of sample-size variation—a dominant yet experimentally tractable source of heterogeneity—before considering other, less controllable factors.

> ***Reviewer #4 (Recommendations for the authors):***
>
> *Other more detailed comments:*
>
> *(1) Introduction*
>
> *L93: When listing reasons why genes do not replicate across different cohorts / datasets, there is also the simple fact that some could be false positives*

We totally agree that some genes may simply represent false-positive findings apart from biological heterogeneity and technical differences between cohorts. Although the MEMORY framework reduces this risk by requiring high reproducibility across 1,000 resamples and multiple sample-size tiers, it cannot eliminate false positives completely. We have added some discussion and explicitly note that external validation in independent datasets is essential for confirming any GEAR before clinical application.

> *(2) Results Section*
>
> *L143: Language like "We also identified the most significant GEARs in individual cancer types" I think is potentially misleading since the "GEAR" lists do not have formal statistical significance attached.*

We removed “significant” ad revised it to “top 1” (Line 115).

> *L153 onward: The pathway analysis results reported do not include any measures of how statistically significant the enrichment was.*

We have now updated the figure legends to clearly indicate that the displayed pathways represent the top significantly enriched results based on adjusted p-values from GO enrichment analyses (Lines 876-878).

> *L168: "A certain degree of correlation with cancer stages (TNM stages) is observed in most cancer types except for COAD, LUSC and PRAD". For statements like this statistical significance should be mentioned in the same sentence or, if these correlations failed to reach significance, that should be explicitly stated.*

In the revised Supplementary Figure 5A–K, we now accompany the visual trends with formal statistical testing. Specifically, for each cancer type, we constructed a contingency table of AJCC stage (I–IV) versus hub-gene subgroup (Low, Mid, High) and applied Pearson's x^2 test (using Monte Carlo approximation with 10^5 replicates if any expected cell count was < 5). The resulting x^2 statistic and p-value are printed beneath each panel. Of the eleven cancer types analyzed, eight showed statistically significant associations ($p < 0.05$), while COAD, LUSC, and PRAD did not. Accordingly, we have make the revision in the manuscript (Line 137139).

> *L171-176: When mentioning which pathways are enriched among the gene lists, please clarify whether these levels of enrichment are statistically significant or not. If the enrichment is significant, please indicate to what degree, and if not I would not mention.*

We agree that the statistical significance of pathway enrichment should be clearly stated and made the revision throughout the manuscript (Line 869, 875, 877).

> *(3) Methods Section*
>
> *L406 - 418: I did not really understand, nor see it explained, what is the motivation and value of cycling through 10%, 20% bootstrapped proportions of patients in the "gradient" approach? I did not see this justified, or motivated by any pre-existing statistical methodology/results. I do not follow the benefit compared to just doing one analysis of all available samples, and using the statistical inference we get "for free" from the survival analysis p-values to quantify sampling uncertainty.*

The ten step-wise sample fractions (10 % to 100 %) allow us to transform each gene's single log-rank P-value into a reproducibility probability: at every fraction we repeat the test 1,000 times and record the proportion of permutations in which the gene is significant. This learning-curve-style resampling not only quantifies how consistently a gene associates with survival under different power conditions but also produces the 0/1 vectors required to compute Survival-Analysis Similarity (SAS) and build the Core Survival Network. A single one-off analysis on the full cohort would yield only one P-value per gene, providing no binary vectors at all—hence no basis for calculating SAS or constructing the network.

> *L417: I assume p < 0.05 in the survival analysis means the nominal p-value, unadjusted for multiple testing. Since we are in the context of many tests please explicitly state if so.*

Yes, $p < 0.05$ refers to the nominal, unadjusted p-value from each log-rank test within a single permutation. In MEMORY these raw p-values are converted immediately into 0/1 "votes" and aggregated over 1 000 permutations and ten sample-size tiers; only the resulting reproducibility probability (A_{ij}) is carried forward. No multiple-testing adjustment is applied at the individual-test level, because a per-iteration FDR or BH step would not materially affect the final A_{ij} ranking. We have revised the manuscript (Line 396)

> *L419-426: I did not see defined what the rows are and what the columns are in the "significant-probability matrix". Are rows genes, columns cancer types? Consequently I was not really sure what actually makes a "GEAR". Is it achieving a significance probability of 0.8 across all 15 cancer subtypes? Or in just one of the tumour datasets?*

In the significant-probability matrix, each row represents a gene, and each column corresponds to a sampling gradient (i.e., increasing sample-size tiers from ~10% to 100%) within a single cancer type. The matrix is constructed independently for each cancer.

GEAR is defined as achieving a significance probability of 0.8 within a single tumor type. Not need to achieve significance probability across all 15 cancer subtypes.

> *L426: The significance probability threshold of 0.8 across 1,000 bootstrapped nominal tests --- used to define the GEAR lists --- has, as far as I can tell, no formal justification. Conceptually, the "significance probability" reflects uncertainty in the patients being used (if I follow their procedure correctly), but as mentioned above, a classical p-value is also designed to reflect sampling uncertainty. So why use the bootstrapping at all?*
>
> *Moreover, the 0.8 threshold is applied on a per-gene basis, so there is no apparent procedure "built in" to adapt to (and account for) different total numbers of genes being tested. Can the authors quantify the false discovery rate associated with this GEAR selection procedure e.g. by running for data with permuted outcome labels? And why do the gradient / bootstrapping at all --- why not just run the nominal survival p-values through a simple Benjamini-Hochberg procedure, and then apply and FDR threshold to define the GEAR lists? Then you would have both multiplicity and error control for the final lists. As it stands, with no form of error control or quantification of noise rates in the GEAR lists I would not recommend promoting their use. There is a long history of variable selection techniques, and various options the authors could have used that would have provided formal error rates for the final GEAR lists (see seminal reviews by eg Heinze et al 2018 Biometrical*
>
> *Journal, or O'Hara and Sillanpaa, 2009, Bayesian Analysis), including, as I say, simple application of a Benjamini-Hochberg to achive multiplicity adjusted FDR control.*

Thank you. We chose the 10 × 1,000 resampling scheme to ask a different question from a single Benjamini–Hochberg scan: does a gene keep re-appearing as significant when cohort composition and statistical power vary from 10 % to 100 % of the data? Converting the 1,000 nominal p-values at each sample fraction into a reproducibility probability A_{ij} allows us to screen for signals that are stable across wide sampling uncertainty rather than relying on one pass through the full cohort. The 0.8 cut-off is an intentionally strict, empirically accepted robustness threshold (analogous to stability-selection); under the global null the chance of exceeding it in 1,000 draws is effectively zero, so the procedure is already highly conservative even before any gene-wise multiplicity correction [1]. Once MEMORY moves beyond this exploratory stage and a final, clinically actionable GEAR catalogue is required, we will add a formal FDR layer after the robustness screen, but for the present proof-of-concept study, we retain the resampling step specifically to capture stability rather than to serve as definitive error control.

> *L427-433: I gathered that SAS reflects, for a particular pair of genes, how likely they are to be jointly significant across bootstraps. If so, perhaps this description or similar could be added since I found a "conceptual" description lacking which would have helped when reading through the maths. Does it make sense to also reflect joint significance across multiple cancer types in the SAS? Or did I miss it and this is already reflected?*

SAS is indeed meant to quantify, within a single cancer type, how consistently two genes are jointly significant across the 1,000 bootstrap resamples performed at a given sample-size tier. In another words, SAS is the empirical probability that the two genes “co-light-up” in the same permutation, providing a measure of shared prognostic behavior beyond what either gene shows alone. We have added this plain language description to the 'Methods' (Lines 405-418).

In the current implementation SAS is calculated separately for each cancer type; it does not aggregate cosignificance across different cancers. Extending SAS to capture joint reproducibility across multiple tumor types is an interesting idea, especially for identifying pan-cancer gene pairs, and we note this as a potential future enhancement of the MEMORY pipeline.

> *L432: "The SAS of significant genes with total genes was calculated, and the significant survival network was constructed" Are the "significant genes" the "GEAR" list extracted above according to the 0.8 threshold? If so, and this is a bit pedantic, I do not think they should be referred to as "significant genes" and that this phrase should be reserved for formal statistical significance.*

We have replaced "significant genes" with "GEAR genes" to avoid any confusion (Lines 421-422).

> *L434: "some SAS values at the top of the rankings were extracted, and the SAS was visualized to a network by Cytoscape. The network was named core survival network (CSN)". I did not see it explicitly stated which nodes actually go into the CSN. The entire GEAR list? What threshold is applied to SAS values in order to determine which edges to include? How was that threshold chosen? Was it data driven? For readers not familiar with what Cytoscape is and how it works could you offer more of an explanation in-text please? I gather it is simply a piece of network visualisation/wrangling software and does not annotate additional information (e.g. external experimental data), which I think is an important point to clarify in the article without needing to look up the reference.*

We have now clarified these points in the revised 'Methods' section, including how the SAS threshold was selected and which nodes were included in the Core Survival Network (CSN). Specifically, the CSN was constructed using the top 1,000 gene pairs with the highest SAS values. This threshold was not determined by a fixed numerical cutoff, but rather chosen empirically after comparing networks built with varying numbers of edges (250, 500, 1,000, 2,000, 6,000, and 8,000; see Reviewer-only Figure 1). We observed that, while increasing the number of edges led to denser networks, the set of hub genes remained largely stable. Therefore, we selected 1,000 edges as a balanced compromise between capturing sufficient biological information and maintaining computational efficiency and interpretability.

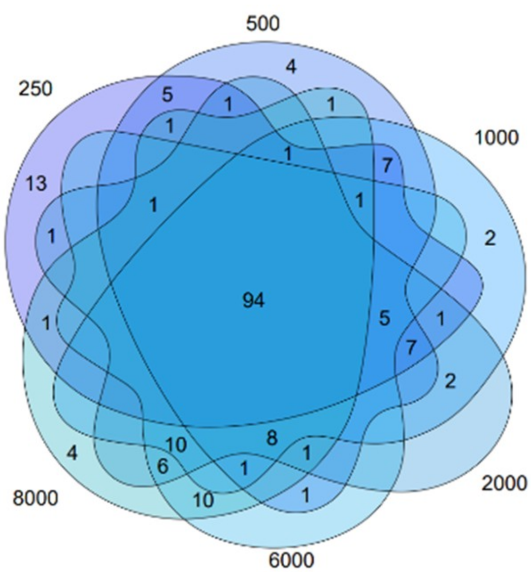
The resulting node list (i.e., the genes present in those top-ranked pairs) is provided in Supplementary Table 4. Cytoscape was used solely as a network visualization platform, and no external annotations or experimental data were added at this stage. We have added a brief clarification in the main text to help readers understand.

> *L437: "The effect of molecular classification by hub genes is indicated that 1000 to 2000 was a range that the result of molecular classification was best." Can you clarify how "best" is assessed here, i.e. by what metric and with which data?*

We apologize for the confusion. Upon constructing the network, we observed that the number of edges affected both the selection of hub genes and the computational complexity. We analyzed the networks with 250, 500, 1,000, 2,000, 6,000 and 8,000 edges, and found that the differences in selected hub genes were small (Author response image 1). Although the networks with fewer edges had lower computational complexity, the choice of 1000 edges was a compromise to the balance between sufficient biological information and manageable computational complexity. Thus, we chose the network with 1,000 edges as it offered a practical balance between computational efficiency and the biological relevance of the hub genes.

Author response image 1.

The intersection of the network constructed by various number of edges.



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